

Performance Analysis

[Reattempt](#)[Download Certificate](#)

Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

Answer Distribution

0

Correct

0

Incorrect

64

Unattempted

Domain Performance

Security, Identity, and Compliance

0%

Application Integration

0%

Databases

0%

Management and Governance

0%

Storage

0%

Networking

0%

Compute

0%

Final Score

**0/64**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes

Question Analysis

✓ Correct	0
✗ Incorrect	0
⊖ Unattempted	64
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Security, Identity, and Compliance



A company recently migrated its applications from an on-premises data center to Amazon EC2 instances. During their time on-premises, these applications suffered a breach due to vulnerabilities in their custom-built code. To prevent future incidents, the company needs a solution that actively scans their EC2 instances for vulnerabilities, generates reports detailing the findings, and allows for automated distribution of those reports. Which AWS service combination best fulfills these requirements?

- ⊗ Deploy AWS Shield to scan the EC2 instances for vulnerabilities. Create an AWS Lambda function to log any findings to AWS CloudTrail.
AWS Shield is designed to protect against DDoS attacks, not to scan for vulnerabilities in EC2 instances. CloudTrail logs API calls, not vulnerability scan results.
- ⊗ Deploy Amazon Macie and AWS Lambda functions to scan the EC2 instances for vulnerabilities. Log any findings to AWS CloudTrail.
Amazon Macie is used for discovering and protecting sensitive data, not for vulnerability scanning of EC2 instances. CloudTrail logs API calls, not vulnerability scan results.
- ⊗ Turn on Amazon GuardDuty. Deploy the GuardDuty agents to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings.
Amazon GuardDuty is a threat detection service that monitors for malicious activity. It doesn't actively scan for vulnerabilities within EC2 instances and doesn't require agents to be deployed on EC2 instances.
- ✔ Turn on Amazon Inspector. Deploy the Amazon Inspector agent to the EC2 instances. Configure an AWS Lambda function to automate the generation and distribution of reports that detail the findings.
Amazon Inspector performs automated security assessments, including vulnerability scanning, on EC2 instances. The Inspector agent is required for these assessments. A Lambda function can automate report generation and distribution.

Overall Explanation:

Amazon Inspector is designed for automated security assessments, including vulnerability scanning, on EC2 instances. It requires the Inspector agent to be deployed on the EC2 instances to perform the scans. Integrating with AWS Lambda allows for the automation of report generation and distribution, meeting the requirement of proactive vulnerability scanning and reporting.

[View Discussions](#)[Ask AI](#)

Question 2

Not Attempted

[Application Integration](#)

A company currently uses an Amazon EC2 instance to run a script that polls and processes messages from an Amazon Simple Queue Service (SQS) queue. The company aims to reduce

operational costs while ensuring that the growing volume of messages in the queue is processed efficiently. What solution should a solutions architect recommend to meet these requirements?

- ☐ Increase the size of the EC2 instance to process messages faster.
Increasing the instance size can process messages faster but won't necessarily reduce operational costs. The EC2 instance will still be running and incurring costs even when the queue is empty or has low message volume.
- ☐ Use Amazon EventBridge to turn off the EC2 instance when the instance is underutilized.
While EventBridge can help manage the EC2 instance, it doesn't eliminate the need for a constantly running instance. There's still operational overhead and costs associated with the EC2 instance.
- ☒ Migrate the script on the EC2 instance to an AWS Lambda function with the appropriate runtime.
Migrating the script to AWS Lambda allows the company to leverage serverless computing. Lambda automatically scales based on the message volume in the SQS queue, and the company only pays for the actual execution time. This reduces operational costs and efficiently handles varying message volumes.
- ☐ Use AWS Systems Manager Run Command to run the script on demand.
While Systems Manager Run Command allows running scripts on demand, it doesn't address the need for continuous polling of the SQS queue. It also introduces additional operational overhead to trigger the script execution.

Overall Explanation:

Migrating the script to AWS Lambda is the most suitable option. Lambda's serverless nature eliminates the need for a dedicated EC2 instance, reducing operational costs. It also scales automatically to handle the growing message volume in the SQS queue, ensuring efficient processing. Lambda costs money only when it's processing, not when idle.

[View Discussions](#)[Ask AI](#)

Question 3

Not Attempted

[Databases](#)

A company uses a legacy application to produce data in CSV format. The legacy application stores the output data in Amazon S3. The company is deploying a new commercial off-the-

shelf (COTS) application that can perform complex SQL queries to analyze data that is stored in Amazon Redshift and Amazon S3 only. However, the COTS application cannot process the .csv files that the legacy application produces. The company cannot update the legacy application to produce data in another format. The company needs to implement a solution so that the COTS application can use the data that the legacy application produces. Which solution will meet these requirements with the LEAST operational overhead?

- ☒ Create an AWS Glue extract, transform, and load (ETL) job that runs on a schedule. Configure the ETL job to process the .csv files and store the processed data in Amazon Redshift.

AWS Glue is a fully managed ETL service that minimizes operational overhead. It can extract data from S3, transform it, and load it into Amazon Redshift. This approach avoids managing infrastructure and simplifies the data transformation process, providing a serverless ETL solution.

- ☐ Develop a Python script that runs on Amazon EC2 instances to convert the .csv files to .sql files. Invoke the Python script on a cron schedule to store the output files in Amazon S3.

Developing and managing a custom Python script on EC2 instances increases operational overhead, including instance management, script maintenance, and potential scaling issues. This solution is more complex than using a managed ETL service.

- ☐ Create an AWS Lambda function and an Amazon DynamoDB table. Use an S3 event to invoke the Lambda function. Configure the Lambda function to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in the DynamoDB table.

Using Lambda and DynamoDB adds unnecessary complexity. The COTS application requires data in Amazon Redshift or S3, but DynamoDB is a NoSQL database. Storing the data in DynamoDB does not directly meet the COTS application's requirements and introduces an additional data store.

- ☐ Use Amazon EventBridge to launch an Amazon EMR cluster on a weekly schedule. Configure the EMR cluster to perform an extract, transform, and load (ETL) job to process the .csv files and store the processed data in an Amazon Redshift table.

Using Amazon EMR for simple CSV transformation is an overkill. EMR is designed for big data processing and involves significant operational overhead, including cluster management, configuration, and cost. It's a less efficient solution compared to AWS Glue.

Overall Explanation:

AWS Glue provides a fully managed, serverless ETL service, making option A the best choice for minimizing operational overhead. It simplifies the process of extracting data

from S3, transforming the CSV data, and loading it into Amazon Redshift, which meets the COTS application's requirements without requiring extensive infrastructure management. Options B, C, and D introduce unnecessary operational overhead or do not directly fulfill the application's data storage requirements.

[View Discussions](#)[Ask AI](#)

Question 4

Not Attempted

[Management and Governance](#)

A company recently migrated its entire IT environment to the AWS Cloud. The company discovers that users are provisioning oversized Amazon EC2 instances and modifying security group rules without using the appropriate change control process. A solutions architect must devise a strategy to track and audit these inventory and configuration changes. Which actions should the solutions architect take to meet these requirements? (Choose two.)

- ☒ **Enable AWS CloudTrail and use it for auditing.**
AWS CloudTrail provides a record of API calls and can be used to audit changes made to EC2 instances and security groups. By analyzing CloudTrail logs, the solutions architect can track who provisioned oversized instances or modified security groups without proper approval.
- ☐ **Use data lifecycle policies for the Amazon EC2 instances.**
Data lifecycle policies are not directly relevant to tracking and auditing inventory and configuration changes.
- ☐ **Enable AWS Trusted Advisor and reference the security dashboard.**
AWS Trusted Advisor is not directly relevant to the requirement of tracking and auditing inventory and configuration changes.
- ☒ **Enable AWS Config and create rules for auditing and compliance purposes.**
AWS Config can record the configuration changes made to resources like EC2 instances and security groups. The solutions architect can create AWS Config rules to monitor for non-compliant changes, like launching certain instance types or opening security group ports without permission. AWS Config would alert on any violations of these rules.
- ☐ **Restore previous resource configurations with an AWS CloudFormation template.**
Restoring previous resource configurations with an AWS CloudFormation template is not directly relevant to the requirement of tracking and auditing inventory and configuration changes.

Overall Explanation:

Enabling AWS CloudTrail allows for auditing of API calls related to resource creation and modification. Enabling AWS Config and creating rules enables continuous monitoring of resource configurations and alerts when deviations from desired states occur. These two services provide the necessary tracking and auditing capabilities to meet the requirements. Options B, C, and E are not relevant to the question.

[View Discussions](#)[Ask AI](#)

Question 5

Not Attempted

Security, Identity, and Compliance



A company has hundreds of Amazon EC2 Linux-based instances in the AWS Cloud. Systems administrators have used shared SSH keys to manage the instances. After a recent audit, the company's security team is mandating the removal of all shared keys. A solutions architect must design a solution that provides secure access to the EC2 instances. Which solution will meet this requirement with the LEAST amount of administrative overhead?

- ☒ Use AWS Systems Manager Session Manager to connect to the EC2 instances.
Systems Manager Session Manager provides secure and auditable instance management without the need for any inbound connections or open ports. It also eliminates the need to manage SSH keys manually and provides a complete audit trail of user activity.
- ☐ Use AWS Security Token Service (AWS STS) to generate one-time SSH keys on demand.
STS can generate temporary IAM credentials or access keys but NOT SSH keys. Generating temporary SSH keys via STS involves more complex configurations and overhead.
- ☐ Allow shared SSH access to a set of bastion instances. Configure all other instances to allow only SSH access from the bastion instances.
This does not remove all shared keys, as requested, and requires managing bastion hosts and SSH access to them.
- ☐ Use an Amazon Cognito custom authorizer to authenticate users. Invoke an AWS Lambda function to generate a temporary SSH key.
Cognito is not meant for internal users, and this solution involves complex setup and higher administrative overhead.

Overall Explanation:

AWS Systems Manager Session Manager is the correct answer because it provides secure shell access to EC2 instances without the need for SSH keys, meeting the security requirement to remove shared SSH keys while minimizing administrative overhead. It eliminates inbound SSH ports and removes manual SSH key management. STS doesn't generate SSH Keys, Bastion Hosts won't fulfill the requirements to remove all shared keys and Cognito is not the appropriate service for this scenario.

[View Discussions](#)[Ask AI](#)

Question 6

Not Attempted

Storage



A development team is tasked with hosting a simple static website for internal team access. The website consists of HTML, CSS, client-side JavaScript, and image files. Considering cost-effectiveness and minimal operational overhead, what is the most suitable AWS service for hosting this website?

- ☐ Containerize the website and host it on AWS Fargate.
Using Fargate involves containerization and adds complexity and costs associated with container management, which is not ideal for a simple static website.
- ☒ Create an Amazon S3 bucket and configure it for static website hosting.
S3 provides a cost-effective, scalable, and durable solution for hosting static websites. It minimizes operational overhead and offers pay-as-you-go pricing.
- ☐ Deploy a web server on an Amazon EC2 instance to host the website.
Using an EC2 instance requires managing the server, which is more expensive and adds operational overhead compared to S3 static website hosting.
- ☐ Configure an Application Load Balancer with an AWS Lambda function using the Express.js framework.
This solution is overly complex and expensive for hosting static content. Lambda and ALB are not optimized for serving static website assets.

Overall Explanation:

For a static website comprising HTML, CSS, JavaScript, and images, Amazon S3 offers the most cost-effective and operationally simple hosting solution. S3 is designed for storing

and serving static content, providing scalability, durability, and a pay-as-you-go pricing model, making it ideal for this scenario. Other options introduce unnecessary complexity and costs.

[View Discussions](#)[Ask AI](#)

Question 7

Not Attempted

Databases



A company is using a fleet of Amazon EC2 instances to ingest data from on-premises data sources. The data is in JSON format, and ingestion rates can be as high as 1 MB/s. When an EC2 instance is rebooted, the data in-flight is lost. The company's data science team wants to query ingested data in near-real time. Which solution provides near-real-time data querying that is scalable with minimal data loss?

- ☒ Publish data to Amazon Kinesis Data Streams. Use Kinesis Data Analytics to query the data.
Publishing data to Amazon Kinesis Data Streams can support ingestion rates as high as 1 MB/s and provide near-real-time data processing. Kinesis Data Analytics can query the ingested data in near-real-time with low latency. This solution also ensures minimal data loss in the event of an EC2 instance reboot since Kinesis Data Streams has a persistent data store.
- ☐ Publish data to Amazon Kinesis Data Firehose with Amazon Redshift as the destination. Use Amazon Redshift to query the data.
While Kinesis Data Firehose and Redshift can be used, the data needs to be loaded into Redshift from Firehose, introducing latency. Redshift also may not be suitable for true 'near real-time' querying as compared to Kinesis Data Analytics.
- ☐ Store ingested data in an EC2 instance store. Publish data to Amazon Kinesis Data Firehose with Amazon S3 as the destination. Use Amazon Athena to query the data.
Storing data in an EC2 instance store is not persistent, and data will be lost during reboots. Athena is also not ideal for near-real-time querying.
- ☐ Store ingested data in an Amazon Elastic Block Store (Amazon EBS) volume. Publish data to Amazon ElastiCache for Redis. Subscribe to the Redis channel to query the data.
While ElastiCache offers fast in-memory storage, it's not designed for high-volume data ingestion like 1 MB/s. It might struggle with scalability and persistence.

Overall Explanation:

The best solution is to publish data to Amazon Kinesis Data Streams and use Kinesis Data Analytics to query the data. This provides near-real-time data ingestion, querying and minimizes data loss with scalability. Kinesis Data Analytics allows for querying data directly from the Kinesis stream with minimal latency.

[View Discussions](#)[Ask AI](#)

Question 8

Not Attempted

Storage



A solutions architect needs to ensure that all objects uploaded to an Amazon S3 bucket are encrypted at rest. Which of the following actions should the architect take to enforce this requirement?

- ☐ Update the bucket policy to deny PutObject requests that do not have the `s3:x-amz-acl` header set.

This option focuses on Access Control Lists (ACLs), which are related to access permissions, not encryption. It won't enforce encryption at rest.

- ☐ Update the bucket policy to deny PutObject requests that do not have the `s3:x-amz-acl` header set to `private`.

This option, similar to the previous one, deals with access control, specifically ensuring that objects are set to private. It does not enforce encryption at rest.

- ☐ Update the bucket policy to deny PutObject requests that do not have the `aws:SecureTransport` header set to `true`.

The `aws:SecureTransport` condition ensures that data is transmitted over HTTPS (in transit encryption), but it doesn't enforce server-side encryption (at rest).

- ☒ Update the bucket policy to deny PutObject requests that do not have the `x-amz-server-side-encryption` header set.

This is the correct approach. The `x-amz-server-side-encryption` header is used to specify the encryption method for objects uploaded to S3. By denying uploads that don't include this header, you force users to explicitly enable encryption during upload.

Overall Explanation:

To ensure all objects in an S3 bucket are encrypted at rest, you must update the bucket policy to deny any PutObject requests lacking the `x-amz-server-side-encryption` header.

This enforces server-side encryption for all uploads. The other options focus on access control or in-transit encryption, not encryption at rest.

[View Discussions](#)[Ask AI](#)

Question 9

Not Attempted

Application Integration



A solutions architect is designing a multi-tier application where users upload images from their mobile devices. The application needs to generate a thumbnail for each image, but the company wants to provide users with immediate confirmation that the image was successfully received, without waiting for the thumbnail generation which can take up to 60 seconds. To accomplish this, the application must asynchronously dispatch requests to different application tiers. How should the solutions architect design the application to meet these requirements?

- ☐ Write a custom AWS Lambda function to generate the thumbnail and alert the user. Use the image upload process as an event source to invoke the Lambda function.

While Lambda can generate thumbnails, this option doesn't guarantee immediate confirmation to the user upon receiving the image.

- ☐ Create an AWS Step Functions workflow. Configure Step Functions to handle the orchestration between the application tiers and alert the user when thumbnail generation is complete.

Step Functions is useful for orchestration, but alerting the user only after thumbnail generation is complete does not satisfy the requirement of providing immediate confirmation upon receiving the image.

- ☒ Create an Amazon Simple Queue Service (Amazon SQS) message queue. As images are uploaded, place a message on the SQS queue for thumbnail generation. Alert the user through an application message that the image was received.

This is the best approach. SQS decouples the image upload and thumbnail generation processes. The application can immediately send a confirmation message to the user after the image is received and the message is placed in the queue, before the thumbnail generation process begins.

- ☐ Create Amazon Simple Notification Service (Amazon SNS) notification topics and subscriptions. Use one subscription with the application to generate the thumbnail after the image upload is complete. Use a second subscription to message the user's mobile app by way of a push notification after thumbnail generation is complete.

SNS is good for fan-out scenarios. However, notifying the user after thumbnail generation doesn't satisfy the requirement of providing immediate feedback upon image reception.

Overall Explanation:

The best solution is to use Amazon SQS to decouple the image upload and thumbnail generation. The application can immediately send a confirmation to the user that the image was received, then queue the thumbnail generation process. This satisfies the requirement of immediate feedback while still generating the thumbnail asynchronously.

[View Discussions](#)[Ask AI](#)

Question 10

Not Attempted

Application Integration



A company's facility uses badge readers at every entrance. When badges are scanned, the readers send a message over HTTPS indicating who attempted to access that entrance. A solutions architect must design a system to process these messages from the sensors. The solution must be highly available, and the results must be readily available for the company's security team to analyze. Which system architecture should the solutions architect recommend?

- ☐ Launch an Amazon EC2 instance to serve as the HTTPS endpoint and process the messages. Configure the EC2 instance to save the results to an Amazon S3 bucket.
Using a single EC2 instance introduces a single point of failure, failing to meet the high availability requirement.
- ☒ Create an HTTPS endpoint in Amazon API Gateway. Configure the API Gateway endpoint to invoke an AWS Lambda function to process the messages and save the results to an Amazon DynamoDB table.
This solution provides a highly available, scalable, and managed architecture. API Gateway handles the HTTPS endpoint, Lambda processes the messages, and DynamoDB provides a durable data store.
- ☐ Use Amazon Route 53 to direct incoming sensor messages to an AWS Lambda function. Configure the Lambda function to process the messages and save the results to an Amazon DynamoDB table.
Exposing Lambda directly to the internet via Route 53 is not a best practice. API Gateway provides features like access control, request validation, and throttling.

- ⊗ Create a gateway VPC endpoint for Amazon S3. Configure a Site-to-Site VPN connection from the facility network to the VPC so that sensor data can be written directly to an S3 bucket by way of the VPC endpoint.

This option adds unnecessary complexity with a VPN and doesn't process the data before storing it. It also lacks scalability and high availability compared to a serverless approach.

Overall Explanation:

The optimal solution is to use Amazon API Gateway as the HTTPS endpoint to receive messages from the badge readers. API Gateway then invokes an AWS Lambda function to process the messages and store the results in an Amazon DynamoDB table. This architecture is highly available, scalable, and provides a managed, serverless approach, best fitting the stated requirements.

 View Discussions

 Ask AI

Question 11

Not Attempted

Storage



A company wants to implement a disaster recovery plan for its primary on-premises file storage volume. The file storage volume is mounted from an Internet Small Computer Systems Interface (iSCSI) device on a local storage server. The file storage volume holds hundreds of terabytes (TB) of data. The company wants to ensure that end users retain immediate access to all file types from the on-premises systems without experiencing latency. Which solution will meet these requirements with the LEAST amount of change to the company's existing infrastructure?

- ⊗ Provision an Amazon S3 File Gateway as a virtual machine (VM) that is hosted on premises. Set the local cache to 10 TB. Modify existing applications to access the files through the NFS protocol. To recover from a disaster, provision an Amazon EC2 instance and mount the S3 bucket that contains the files.

This option requires application changes and the limited local cache will not provide immediate access to all files.

- ⊗ Provision an AWS Storage Gateway tape gateway. Use a data backup solution to back up all existing data to a virtual tape library. Configure the data backup solution to run nightly after the initial backup is complete. To recover from a disaster, provision an Amazon EC2 instance and restore the data to an Amazon Elastic Block Store (Amazon EBS) volume from the volumes in the virtual tape library.

This option involves backing up to a virtual tape library and restoring during disaster recovery which will not provide immediate access and involves a restore process.

- ⊗ Provision an AWS Storage Gateway Volume Gateway cached volume. Set the local cache to 10 TB. Mount the Volume Gateway cached volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance.

This option uses a cached volume which only stores frequently accessed data locally and will not provide immediate access to all files without latency.

- ✓ Provision an AWS Storage Gateway Volume Gateway stored volume with the same amount of disk space as the existing file storage volume. Mount the Volume Gateway stored volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance.

This option uses a stored volume which retains all data locally, providing immediate access with low latency. Scheduled snapshots provide a mechanism for disaster recovery.

Overall Explanation:

The best solution is to use AWS Storage Gateway Volume Gateway stored volume. Stored volumes retain the entire dataset locally, allowing immediate access with low latency. The other options either don't provide immediate access, require application changes, or do not retain all data locally. Therefore, option D is the most suitable solution that meets the stated requirements.

[View Discussions](#)[Ask AI](#)

Question 12

Not Attempted

Security, Identity, and Compliance



A company hosts a web application from an Amazon S3 bucket. The application uses Amazon Cognito as an identity provider to authenticate users and return a JSON Web Token (JWT) that provides access to protected resources stored in another S3 bucket. Upon deployment, users report errors and cannot access the protected content. A solutions architect must resolve this

issue by providing proper permissions so that users can access the protected content. Which solution meets these requirements?

- ☒ Update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content.

Updating the Cognito identity pool to assume the proper IAM role grants authenticated users the necessary permissions to access the protected content in the S3 bucket. IAM roles provide temporary, limited-privilege credentials to access AWS resources.

- ☐ Update the S3 ACL to allow the application to access the protected content.

Updating the S3 ACL is not the ideal solution. While it might grant access, it's generally better to manage access through IAM roles assumed by Cognito, rather than relying on ACLs for authentication.

- ☐ Redeploy the application to Amazon S3 to prevent eventually consistent reads in the S3 bucket from affecting the ability of users to access the protected content.

Redeploying the application to S3 will not solve permission problems. Eventual consistency is also unlikely to be the root cause of the described issue.

- ☐ Update the Amazon Cognito pool to use custom attribute mappings within the identity pool and grant users the proper permissions to access the protected content.

While custom attribute mappings can be used to fine-tune permissions, the core issue is likely that the Cognito identity pool doesn't have the necessary IAM role to allow access to the protected S3 bucket. Updating the IAM role associated with the identity pool should be performed first.

Overall Explanation:

The correct answer is to update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content. Amazon Cognito identity pools assign temporary, limited-privilege credentials to authenticated users to access AWS resources. The permissions for each user are controlled through IAM roles that you create. This approach aligns with AWS best practices for managing permissions and access control.

 View Discussions

 Ask AI

Question 13

Not Attempted

Storage



An image hosting company uploads large assets to Amazon S3 Standard buckets. The company uses multipart upload in parallel through S3 APIs and overwrites existing objects if the same

object is uploaded again. For the first 30 days after upload, the objects are accessed frequently. After 30 days, object access patterns become inconsistent. The company must optimize its S3 storage costs while maintaining high availability and resiliency. Which combination of actions should a solutions architect recommend to meet these requirements? (Choose two.)

- ☒ Move assets to S3 Intelligent-Tiering after 30 days.
S3 Intelligent-Tiering automatically moves infrequently accessed objects to lower-cost tiers based on access patterns, optimizing costs for unpredictable access patterns.
- ☒ Configure an S3 Lifecycle policy to clean up incomplete multipart uploads.
Incomplete multipart uploads consume storage space and increase costs. An S3 Lifecycle policy can automatically remove these incomplete uploads.
- ☐ Configure an S3 Lifecycle policy to clean up expired object delete markers.
Expired object delete markers do not contribute significantly to cost optimization in this scenario.
- ☐ Move assets to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.
While S3 Standard-IA is cheaper, S3 Intelligent-Tiering is better suited for inconsistent access patterns, potentially moving objects to even lower-cost tiers like Glacier Instant Retrieval if they are rarely accessed.
- ☐ Move assets to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.
S3 One Zone-IA does not provide the same level of high availability and resiliency as S3 Standard or S3 Intelligent-Tiering.

Overall Explanation:

To optimize S3 storage costs while maintaining high availability and resiliency, the solutions architect should recommend the following actions: Move assets to S3 Intelligent-Tiering after 30 days to take advantage of automatic tiering based on access patterns. Configure an S3 Lifecycle policy to clean up incomplete multipart uploads to prevent wasted storage space and costs. S3 Intelligent-Tiering handles the inconsistent access patterns by moving data to the most cost effective tier. Cleaning up incomplete multipart uploads is a standard best practice for cost optimization.

[View Discussions](#)[Ask AI](#)

Question 14

Not Attempted

Security, Identity, and Compliance



A solutions architect must secure a VPC network that hosts Amazon EC2 instances. The EC2 instances contain highly sensitive data and run in a private subnet. According to company policy, the EC2 instances that run in the VPC can access only approved third-party software repositories on the internet for software product updates that use the third party's URL. Other internet traffic must be blocked. Which solution meets these requirements?

- ☒ Update the route table for the private subnet to route the outbound traffic to an AWS Network Firewall firewall. Configure domain list rule groups.

AWS Network Firewall allows for filtering traffic based on domain names or URLs, meeting the requirement of allowing access only to approved third-party software repositories. The domain list rule groups can be configured to only allow the specified URLs, and block the remaining traffic.

- ☐ Set up an AWS WAF web ACL. Create a custom set of rules that filter traffic requests based on source and destination IP address range sets.

AWS WAF primarily protects against web application attacks and works at Layer 7. It does not provide functionality to filter outbound traffic based on URLs for EC2 instances.

- ☐ Implement strict inbound security group rules. Configure an outbound rule that allows traffic only to the authorized software repositories on the internet by specifying the URLs.

Security Groups operate at the transport layer (Layer 4) and cannot filter traffic based on URLs. They can only control traffic based on IP addresses, ports, and protocols.

- ☐ Configure an Application Load Balancer (ALB) in front of the EC2 instances. Direct all outbound traffic to the ALB. Use a URL-based rule listener in the ALB's target group for outbound access to the internet.

An ALB is designed for load balancing inbound HTTP/HTTPS traffic. It is not designed to manage or filter outbound traffic from EC2 instances to the internet.

Overall Explanation:

The correct answer is A. AWS Network Firewall allows for granular control over network traffic, including the ability to filter based on domain names and URLs. Configuring domain list rule groups enables the solution architect to allow outbound traffic only to the approved third-party software repositories while blocking all other internet traffic. Security Groups cannot filter based on URLs, and AWS WAF is designed for inbound web traffic, not outbound EC2 traffic. An ALB is for load balancing inbound requests and not for filtering

outbound traffic. Therefore, AWS Network Firewall is the appropriate solution to meet the given requirements.

[View Discussions](#)[Ask AI](#)

Question 15

Not Attempted

Application Integration



A company hosts a three-tier e-commerce application on AWS, with the website on Amazon S3 integrated with an API on three EC2 instances behind an Application Load Balancer (ALB). The API handles sales requests asynchronously using backend workers. During product launches, the company anticipates a significant surge in sales requests. To ensure all requests are successfully processed during these peak events, what should a solutions architect recommend?

- ☐ Add an Amazon CloudFront distribution for the dynamic content. Increase the number of EC2 instances to handle the increase in traffic.

Adding CloudFront for dynamic content and simply increasing EC2 instances may not guarantee that all requests will be processed successfully, especially during sudden spikes. This approach might not scale quickly enough to handle the load and could lead to dropped requests.

- ☐ Add an Amazon CloudFront distribution for the static content. Place the EC2 instances in an Auto Scaling group to launch new instances based on network traffic.

While CloudFront for static content improves website performance and Auto Scaling helps scale the API tier, it doesn't guarantee that all requests will be processed successfully. The auto-scaling process might not react quickly enough to prevent some requests from being dropped during sudden spikes.

- ☐ Add an Amazon CloudFront distribution for the dynamic content. Add an Amazon ElastiCache instance in front of the ALB to reduce traffic for the API to handle.

Caching dynamic content may not be appropriate and adding ElastiCache in front of the ALB would not ensure all sales requests are processed successfully, requests may still get dropped during the spikes.

- ☒ Add an Amazon CloudFront distribution for the static content. Add an Amazon Simple Queue Service (Amazon SQS) queue to receive requests from the website for later processing by the EC2 instances.

This option is the most suitable. CloudFront for static content optimizes website performance. SQS decouples the web tier from the API tier, ensuring all sales requests are reliably stored in the queue,

even during peak events. The backend workers can then process the requests asynchronously from the queue, guaranteeing that no requests are lost.

Overall Explanation:

The optimal solution involves using CloudFront for static content and Amazon SQS to handle sales requests. CloudFront enhances website performance, while SQS acts as a buffer, reliably storing requests and ensuring that all requests are processed asynchronously by the backend workers, even during sudden surges in traffic. This decoupling via SQS guarantees no request loss and successful processing. Auto scaling alone (Option B) does not guarantee that all requests will be processed successfully as auto scaling may not react quickly enough during sudden spikes.

[View Discussions](#)[Ask AI](#)

Question 16

Not Attempted

Security, Identity, and Compliance



A security audit reveals that Amazon EC2 instances are not being patched regularly. A solutions architect needs to provide a solution that will run regular security scans across a large fleet of EC2 instances. The solution should also patch the EC2 instances on a regular schedule and provide a report of each instance's patch status.

- ☒ Set up Amazon Macie to scan the EC2 instances for software vulnerabilities. Set up a cron job on each EC2 instance to patch the instance on a regular schedule.
Amazon Macie is a data security and data privacy service that discovers, classifies, and protects sensitive data stored in Amazon S3. It is not designed to scan EC2 instances for software vulnerabilities or patch them.
- ☒ Turn on Amazon GuardDuty in the account. Configure GuardDuty to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Session Manager to patch the EC2 instances on a regular schedule.
Amazon GuardDuty is a threat detection service that continuously monitors your AWS accounts and workloads for malicious activity and delivers security findings. It does not patch EC2 instances, and while Session Manager provides secure instance access, it doesn't automate patching.
- ☒ Set up Amazon Detective to scan the EC2 instances for software vulnerabilities. Set up an Amazon EventBridge scheduled rule to patch the EC2 instances on a regular schedule.

Amazon Detective analyzes, investigates, and quickly identifies the root cause of suspicious activities or security findings. It is not designed to scan for software vulnerabilities or manage patching.

- ✔ Turn on Amazon Inspector in the account. Configure Amazon Inspector to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Patch Manager to patch the EC2 instances on a regular schedule.

Amazon Inspector is a security assessment service that automatically assesses applications for vulnerabilities or deviations from best practices. AWS Systems Manager Patch Manager automates patching and provides reporting on patch status. This option provides a complete solution.

Overall Explanation:

Amazon Inspector scans EC2 instances for software vulnerabilities, while AWS Systems Manager Patch Manager automates the patching process and provides reports on patch status, thus meeting all the requirements. Therefore, option D is the correct solution.

 View Discussions

 Ask AI

Question 17

Not Attempted

Application Integration



A company runs an online marketplace web application on AWS. The application serves hundreds of thousands of users during peak hours. The company needs a scalable, near-real-time solution to share the details of millions of financial transactions with several other internal applications. Transactions also need to be processed to remove sensitive data before being stored in a document database for low-latency retrieval. Which solution should a solutions architect recommend to meet these requirements?

- ✘ Store the transactions data into Amazon DynamoDB. Set up a rule in DynamoDB to remove sensitive data from every transaction upon write. Use DynamoDB Streams to share the transactions data with other applications.

DynamoDB streams are not suitable for real-time sharing. Furthermore, directly modifying data within DynamoDB might impact performance.

- ⊗ Stream the transactions data into Amazon Kinesis Data Firehose to store data in Amazon DynamoDB and Amazon S3. Use AWS Lambda integration with Kinesis Data Firehose to remove sensitive data. Other applications can consume the data stored in Amazon S3.

Kinesis Data Firehose doesn't directly support DynamoDB as a destination. Storing in S3 doesn't fulfill the requirement of a document database for low-latency retrieval.

- ✔ Stream the transactions data into Amazon Kinesis Data Streams. Use AWS Lambda integration to remove sensitive data from every transaction and then store the transactions data in Amazon DynamoDB. Other applications can consume the transactions data off the Kinesis data stream.

This solution uses Kinesis Data Streams for scalable real-time data ingestion, Lambda for data transformation and sensitive data removal, and DynamoDB for low-latency storage and retrieval. Other applications can then consume data from DynamoDB.

- ⊗ Store the batched transactions data in Amazon S3 as files. Use AWS Lambda to process every file and remove sensitive data before updating the files in Amazon S3. The Lambda function then stores the data in Amazon DynamoDB. Other applications can consume transaction files stored in Amazon S3.

Batch processing with S3 and Lambda does not meet the near-real-time requirement. S3 isn't a suitable document database for low-latency retrieval either.

Overall Explanation:

The optimal solution is to use Amazon Kinesis Data Streams to ingest the transaction data in near-real-time. AWS Lambda can be integrated to process each transaction, removing sensitive information. Processed data is then stored in Amazon DynamoDB for low-latency retrieval. Other applications can consume the transactions data off the Kinesis data stream. Therefore, option C is the correct answer. Options A, B, and D are not appropriate because they do not fully meet the requirements of near-real-time processing, data transformation, low-latency retrieval from a document database, and scalable data sharing.

 View Discussions

 Ask AI

Question 18

Not Attempted

Databases



A company is planning to store data on Amazon RDS DB instances. The company must encrypt the data at rest. What should a solutions architect do to meet this requirement?

- ✔ Create a key in AWS Key Management Service (AWS KMS). Enable encryption for the DB instances.

This is the correct approach. Amazon RDS encryption at rest is enabled using keys managed by AWS KMS. You can create a key in KMS and then enable encryption on the RDS instance using that key.

- ✘ Create an encryption key. Store the key in AWS Secrets Manager. Use the key to encrypt the DB instances.

Secrets Manager is used for storing secrets like database credentials, not for directly managing encryption keys for RDS at-rest encryption.

- ✘ Generate a certificate in AWS Certificate Manager (ACM). Enable SSL/TLS on the DB instances by using the certificate.

ACM certificates are used for encrypting data in transit (SSL/TLS), not data at rest.

- ✘ Generate a certificate in AWS Identity and Access Management (IAM). Enable SSL/TLS on the DB instances by using the certificate.

IAM is used for managing permissions, and while certificates can be uploaded to IAM, ACM is the correct service for SSL/TLS certificates. Also, SSL/TLS encrypts data in transit, not at rest.

Overall Explanation:

To encrypt data at rest in Amazon RDS, you should create a key in AWS Key Management Service (AWS KMS) and then enable encryption for the DB instances. This uses the RDS encryption feature that integrates with AWS KMS to encrypt the underlying storage, automated backups, read replicas, and snapshots. Options involving Secrets Manager or ACM certificates are for different purposes (credential management and encrypting data in transit, respectively).

 View Discussions

 Ask AI

Question 19

Not Attempted

Networking



A company needs to migrate 20 TB of data from its data center to the AWS Cloud within 30 days. The company's network bandwidth is limited to 15 Mbps and cannot exceed 70% utilization. What should a solutions architect do to meet these requirements?

- ✔ Use AWS Snowball.

AWS Snowball is designed for transferring large amounts of data when network bandwidth is limited. Given the limited bandwidth and the large data volume, Snowball provides a faster and more reliable solution than transferring over the network. It allows the data to be physically shipped to AWS.

⊗ Use AWS DataSync.

AWS DataSync is a data transfer service that automates and accelerates moving data between on-premises storage and Amazon S3, Amazon EFS, or Amazon FSx for Windows File Server. While DataSync is useful for ongoing replication, the limited bandwidth in this scenario makes it less suitable for the initial 20 TB migration within the 30-day timeframe.

⊗ Use a secure VPN connection.

Using a secure VPN connection alone will not solve the bandwidth limitations. A VPN provides secure data transfer but does not increase the available bandwidth. The limited bandwidth would still make it impossible to transfer 20 TB of data in 30 days.

⊗ Use Amazon S3 Transfer Acceleration.

Amazon S3 Transfer Acceleration utilizes the CloudFront network to accelerate data transfers to S3. However, it relies on network bandwidth and is not effective when bandwidth is severely limited as in this scenario. It will not overcome the fundamental limitation of the 15 Mbps connection with 70% utilization.

Overall Explanation:

Given the 15 Mbps network bandwidth limitation with a maximum 70% utilization, the effective bandwidth is approximately 1.31 MB/s. Transferring 20 TB of data at this rate would take approximately 193 days, far exceeding the 30-day requirement. AWS Snowball provides a physical storage device that can be shipped to the data center, loaded with the data, and then shipped back to AWS. This allows for much faster data transfer compared to relying solely on the limited network bandwidth.

 View Discussions

 Ask AI

Question 20

Not Attempted

Security, Identity, and Compliance



A company needs to provide its employees with secure access to confidential and sensitive files. The company wants to ensure that the files can be accessed only by authorized users, and the files must be downloaded securely to the employees' devices. The files are currently stored

on an on-premises Windows file server, which is running out of capacity due to an increase in remote usage. Which solution will meet these requirements?

- ☐ **Migrate the file server to an Amazon EC2 instance in a public subnet. Configure the security group to limit inbound traffic to the employees' IP addresses.**
This solution does not provide adequate security for accessing confidential files, and managing IP address whitelists for remote users can be cumbersome and unreliable.
- ☒ **Migrate the files to an Amazon FSx for Windows File Server file system. Integrate the Amazon FSx file system with the on-premises Active Directory. Configure AWS Client VPN.**
This solution provides a scalable, fully managed Windows file storage solution accessible from anywhere. Integrating with on-premises Active Directory ensures centralized access control. AWS Client VPN provides a secure and managed VPN solution that allows employees to access files securely.
- ☐ **Migrate the files to Amazon S3, and create a private VPC endpoint. Create a signed URL to allow download.**
Signed URLs provide temporary access, but anyone with the URL can download the file during its validity period, which might not be suitable for highly sensitive data. This option doesn't guarantee secure downloads to devices.
- ☐ **Migrate the files to Amazon S3, and create a public VPC endpoint. Allow employees to sign on with AWS IAM Identity Center (AWS Single Sign-On).**
Using a public VPC endpoint exposes S3 to the internet, which is not secure. While AWS IAM Identity Center handles authentication, it doesn't guarantee secure downloading of the files to the employees' devices.

Overall Explanation:

Migrating the files to Amazon FSx for Windows File Server integrated with on-premises Active Directory, combined with AWS Client VPN, provides a scalable, secure, and manageable solution. FSx addresses the capacity issues, Active Directory integration ensures authorized access, and the VPN ensures secure data transfer. Therefore option B is the best solution.

[View Discussions](#)[Ask AI](#)

A company's application runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. On the first day of every month at midnight, the application becomes much slower when the month-end financial calculation batch runs. This causes the CPU utilization of the EC2 instances to immediately peak to 100%, which disrupts the application. What should a solutions architect recommend to ensure the application is able to handle the workload and avoid downtime?

- ☐ Configure an Amazon CloudFront distribution in front of the ALB.

Using CloudFront will not prevent the CPU spike on the EC2 instances caused by the month-end calculations.

- ☐ Configure an EC2 Auto Scaling simple scaling policy based on CPU utilization.

A simple scaling policy based on CPU utilization would reactively scale, but the CPU is already at 100% which means the application is already disrupted. Also it takes time for new instances to launch.

- ☒ Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule.

By configuring a scheduled scaling policy, the EC2 Auto Scaling group can proactively launch additional EC2 instances before the CPU utilization peaks to 100%. This will ensure that the application can handle the workload during the month-end financial calculation batch, and avoid any disruption or downtime.

- ☐ Configure Amazon ElastiCache to remove some of the workload from the EC2 instances.

ElastiCache helps with caching frequently accessed data. It does not directly address the issue of the monthly peak workload caused by the financial calculations.

Overall Explanation:

The most appropriate solution to handle the increased workload during the monthly batch run and avoid downtime would be to configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule. This proactively scales the EC2 instances before the monthly batch run begins, ensuring the application handles the increased workload without downtime. A scheduled scaling policy can increase the number of instances in the Auto Scaling group before the batch run and decrease the number of instances after the

batch run is complete. This ensures resources are available when needed and not wasted when not needed.

[View Discussions](#)[Ask AI](#)

Question 22

Not Attempted

Security, Identity, and Compliance



A company needs to provide its customers with the ability to download files stored in Amazon S3 using their existing on-premises Microsoft Active Directory credentials. The customer's application utilizes an SFTP client for file downloads. Which solution meets these requirements with the LEAST operational overhead and without requiring any modifications to the customer's application?

- ☒ Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication.
AWS Transfer Family directly supports SFTP and integrates with Active Directory, minimizing operational overhead and requiring no changes to the client application.
- ☐ Set up AWS Database Migration Service (AWS DMS) to synchronize the on-premises client with Amazon S3. Configure integrated Active Directory authentication.
AWS DMS is designed for database migrations, not file transfers, and would require significant changes and overhead.
- ☐ Set up AWS DataSync to synchronize between the on-premises location and the S3 location by using AWS IAM Identity Center (AWS Single Sign-On).
AWS DataSync is for data transfer and not SFTP. Integrating with IAM Identity Center does not directly provide SFTP access for existing Active Directory users without application changes.
- ☐ Set up a Windows Amazon EC2 instance with SFTP to connect the on-premises client with Amazon S3. Integrate AWS Identity and Access Management (IAM).
Using an EC2 instance involves managing an operating system and SFTP server, which significantly increases operational overhead. Directly integrating IAM would also require application changes.

Overall Explanation:

AWS Transfer Family is the best solution because it's a fully managed service specifically designed for transferring files via SFTP to and from Amazon S3. It integrates seamlessly with Active Directory for authentication, minimizing operational overhead and eliminating

the need for application changes. Options B, C, and D involve more complex configurations and potential application modifications, making them less suitable.

[View Discussions](#)[Ask AI](#)

Question 23

Not Attempted

[Compute](#)

A company anticipates a surge in demand for its application. To handle this, they need to rapidly provision large Amazon EC2 instances from a custom Amazon Machine Image (AMI). These instances will be managed within an Auto Scaling group. The key requirement is minimizing the instance initialization latency to ensure the application can quickly scale and meet the demand. Which solution provides the fastest initialization for new EC2 instances?

- ☐ Use the `aws ec2 register-image` command to create an AMI from a snapshot. Use AWS Step Functions to replace the AMI in the Auto Scaling group.

This option focuses on creating an AMI but doesn't address the initialization latency issue specifically. Registering an image doesn't pre-warm the underlying EBS volumes.

- ☒ Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI.

Enabling EBS fast snapshot restore pre-warms the EBS volumes, significantly reducing the time it takes to create volumes and launch instances from them. This minimizes initialization latency.

- ☐ Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager (Amazon DLM). Create an AWS Lambda function that modifies the AMI in the Auto Scaling group.

Amazon DLM helps with AMI management but doesn't directly address the speed of instance initialization. Lambda functions can automate tasks but won't pre-warm EBS volumes.

- ☐ Use Amazon EventBridge to invoke AWS Backup lifecycle policies that provision AMIs. Configure Auto Scaling group capacity limits as an event source in EventBridge.

This option involves AWS Backup and EventBridge, which are useful for backup and event-driven automation, but it doesn't focus on minimizing instance initialization latency.

Overall Explanation:

The correct answer is to enable Amazon EBS fast snapshot restore on a snapshot and then create the AMI. EBS fast snapshot restore minimizes initialization latency by pre-warming the EBS volumes associated with the snapshot. This ensures that new instances launched from the AMI can quickly start serving traffic, meeting the demand surge.

[View Discussions](#)[Ask AI](#)

Question 24

Not Attempted

Security, Identity, and Compliance



A company hosts a multi-tier web application that uses an Amazon Aurora MySQL DB cluster for storage. The application tier is hosted on Amazon EC2 instances. The company's IT security guidelines mandate that the database credentials be encrypted and rotated every 14 days. What should a solutions architect do to meet this requirement with the LEAST operational effort?

- ☒ Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days.
AWS Secrets Manager allows for automated rotation of credentials. It can integrate with KMS to encrypt secrets and can be configured to rotate credentials on a schedule, minimizing operational overhead.
- ☐ Create two parameters in AWS Systems Manager Parameter Store: one for the user name as a string parameter and one that uses the SecureString type for the password. Select AWS Key Management Service (AWS KMS) encryption for the password parameter, and load these parameters in the application tier. Implement an AWS Lambda function that rotates the password every 14 days.
While Systems Manager Parameter Store can store encrypted secrets, it does not natively support automatic rotation. Implementing a Lambda function adds unnecessary operational complexity.
- ☐ Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system in all EC2 instances of the application tier. Restrict the access to the file on the file system so that the application can read the file and that only super users can modify the file. Implement an AWS Lambda function that rotates the key in Aurora every 14 days and writes new credentials into the file.

Storing credentials in EFS and using a Lambda function for rotation adds significant operational overhead compared to a managed service like Secrets Manager.

- ⊗ Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon S3 bucket that the application uses to load the credentials. Download the file to the application regularly to ensure that the correct credentials are used. Implement an AWS Lambda function that rotates the Aurora credentials every 14 days and uploads these credentials to the file in the S3 bucket.

Storing credentials in S3 and using a Lambda function for rotation introduces unnecessary complexity compared to using AWS Secrets Manager. It also involves managing file downloads and synchronization.

Overall Explanation:

AWS Secrets Manager provides a managed service for storing and rotating database credentials. By creating a secret with the Aurora DB cluster credentials, encrypting it with KMS, and configuring a 14-day rotation period, the requirement is met with minimal operational effort. Parameter Store does not natively handle password rotation. EFS and S3 involve unnecessary complexity in managing credentials and rotation with Lambda functions. Therefore, using AWS Secrets Manager offers the least operational overhead.

 View Discussions

 Ask AI

Question 25

Not Attempted

Databases



A company hosts a web application on AWS with an Amazon RDS for MySQL database consisting of a primary instance and five read replicas for scaling. The read replicas must have a maximum lag of 1 second behind the primary instance. The database regularly runs scheduled stored procedures. As website traffic increases, the replicas experience greater lag during peak load periods. A solutions architect needs to minimize the replication lag, application code changes, and ongoing operational overhead. Which solution best meets these requirements?

- ✔ Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions.

Migrating to Aurora MySQL offers improved replication performance and scalability compared to RDS for MySQL. Aurora Replicas provide faster replication, reducing lag, while Auto Scaling adjusts the number of replicas based on traffic. Replacing stored procedures with Aurora MySQL native

functions can further improve performance. This minimizes application changes and ongoing overhead.

- ⊗ Deploy an Amazon ElastiCache for Redis cluster in front of the database. Modify the application to check the cache before the application queries the database. Replace the stored procedures with AWS Lambda functions.

Adding ElastiCache does not directly address replication lag. It also requires significant changes to the application code to utilize the cache effectively. Replacing stored procedures with Lambda functions adds complexity and may not improve replication lag.

- ⊗ Migrate the database to a MySQL database that runs on Amazon EC2 instances. Choose large, compute optimized EC2 instances for all replica nodes. Maintain the stored procedures on the EC2 instances.

Moving to EC2 instances from RDS adds operational overhead, as you are responsible for managing the database. This option also doesn't inherently solve the replication lag issue and doesn't minimize ongoing operational overhead.

- ⊗ Migrate the database to Amazon DynamoDB. Provision a large number of read capacity units (RCUs) to support the required throughput, and configure on-demand capacity scaling. Replace the stored procedures with DynamoDB streams.

Migrating to DynamoDB, a NoSQL database, would require extensive changes to the application code and logic. DynamoDB streams are not a direct replacement for stored procedures in a relational database context.

Overall Explanation:

Migrating to Amazon Aurora MySQL (Option A) provides the best solution for reducing replication lag while minimizing application changes and operational overhead. Aurora Replicas offer faster replication compared to standard RDS read replicas, and Aurora Auto Scaling dynamically adjusts the number of replicas to handle varying workloads. Replacing stored procedures with Aurora native functions further optimizes performance. The other options involve either significant application code changes or increased operational complexity without directly addressing the replication lag issue as effectively.

[View Discussions](#)[Ask AI](#)

Question 26

Not Attempted

[Databases](#)

A solutions architect must create a disaster recovery (DR) plan for a high-volume software as a service (SaaS) platform. All data for the platform is stored in an Amazon Aurora MySQL DB cluster. The DR plan must replicate data to a secondary AWS Region. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Use MySQL binary log replication to an Aurora cluster in the secondary Region. Provision one DB instance for the Aurora cluster in the secondary Region.

MySQL binary log replication requires enabling binary logging and managing the replication process. This can be complex and might not be the most cost-effective solution compared to Aurora's built-in features.

- ✔ Set up an Aurora global database for the DB cluster. When setup is complete, remove the DB instance from the secondary Region.

Setting up an Aurora global database and then removing the DB instance in the secondary region creates a 'headless' secondary cluster. This configuration leverages Aurora's storage replication without the cost of a running DB instance, making it a very cost-effective DR solution.

- ⊗ Use AWS Database Migration Service (AWS DMS) to continuously replicate data to an Aurora cluster in the secondary Region. Remove the DB instance from the secondary Region.

AWS DMS is generally used for migrating databases between different database engines or versions. While it can replicate data, it's not the most efficient or cost-effective solution for replicating within Aurora, especially for high-volume data.

- ⊗ Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region.

Setting up an Aurora global database with a DB instance in the secondary Region provides faster failover capabilities but will incur costs for the running instance, making it less cost-effective than a headless cluster during normal operation.

Overall Explanation:

The most cost-effective solution is to set up an Aurora global database and then remove the DB instance from the secondary Region, creating a headless cluster. This approach utilizes Aurora's storage replication capabilities without incurring the compute costs of a running DB instance in the secondary region. This provides a cost-optimized DR solution while maintaining data replication.

 View Discussions

 Ask AI

Question 27

Not Attempted

Management and Governance



A company hosts its multi-tier applications on AWS. For compliance, governance, auditing, and security, the company must track configuration changes on its AWS resources and record a history of API calls made to these resources. What should a solutions architect do to meet these requirements?

- ☐ Use AWS CloudTrail to track configuration changes and AWS Config to record API calls.

AWS CloudTrail is designed for recording API calls, not configuration changes. AWS Config is designed for configuration tracking.

- ☒ Use AWS Config to track configuration changes and AWS CloudTrail to record API calls.

AWS Config tracks resource configuration changes, providing a history for auditing and compliance. AWS CloudTrail captures API calls, providing a record of actions taken on AWS resources.

- ☐ Use AWS Config to track configuration changes and Amazon CloudWatch to record API calls.

Amazon CloudWatch is primarily for monitoring metrics and logs, not for comprehensively recording API calls for auditing purposes.

- ☐ Use AWS CloudTrail to track configuration changes and Amazon CloudWatch to record API calls.

AWS CloudTrail records API calls, not configuration changes. CloudWatch is for monitoring, not API call recording.

Overall Explanation:

AWS Config is the appropriate service for tracking changes to AWS resource configurations and providing a history of those changes for compliance and auditing. AWS CloudTrail is specifically designed to record API calls made to AWS resources, capturing details such as who made the call, when, and what resources were affected. Therefore, the correct answer is to use AWS Config for configuration changes and AWS CloudTrail for recording API calls.

[View Discussions](#)[Ask AI](#)

Question 28

Not Attempted

Security, Identity, and Compliance



A media company hosts its website on AWS. The website application's architecture includes a fleet of Amazon EC2 instances behind an Application Load Balancer (ALB) and a database that is hosted on Amazon Aurora. The company's cybersecurity team reports that the application is vulnerable to SQL injection. How should the company resolve this issue?

- ☒ Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF.

AWS WAF is a web application firewall that helps protect your web applications from common web exploits, like SQL injection, that could affect application availability, compromise security, or consume excessive resources. It can be configured with rules to block SQL injection patterns.

- ☐ Create an ALB listener rule to reply to SQL injections with a fixed response.

While an ALB can route traffic based on rules, it's not designed to specifically identify and block SQL injection attacks. Returning a fixed response doesn't address the underlying vulnerability.

- ☐ Subscribe to AWS Shield Advanced to block all SQL injection attempts automatically.

AWS Shield Advanced primarily protects against DDoS attacks. While it provides some WAF capabilities, it's not the primary service for addressing SQL injection vulnerabilities. AWS WAF is better suited for this.

- ☐ Set up Amazon Inspector to block all SQL injection attempts automatically.

Amazon Inspector is a vulnerability management service that scans EC2 instances and other AWS resources for software vulnerabilities and unintended network exposure. It doesn't actively block SQL injection attempts in real-time.

Overall Explanation:

AWS WAF is the correct choice because it's designed to protect web applications from common web exploits, including SQL injection. By placing AWS WAF in front of the ALB and configuring the appropriate web ACLs, the company can inspect incoming traffic and block requests that match SQL injection patterns. AWS Shield primarily focuses on DDoS protection, and Amazon Inspector is for vulnerability assessments, not real-time attack blocking. ALB listener rules are also not equipped for this functionality.

 View Discussions

 Ask AI

Question 29

Not Attempted

Databases



A company has an Amazon S3 data lake that is governed by AWS Lake Formation. The company needs to create visualizations in Amazon QuickSight by joining the data in the data lake with operational data stored in an Amazon Aurora MySQL database. The company must enforce column-level authorization, ensuring the marketing team can access only a specific subset of columns within the database. Which solution satisfies these requirements with the LEAST operational overhead?

- ⊗ Use Amazon EMR to ingest the data directly from the database to the QuickSight SPICE engine. Include only the required columns.

This option involves managing an EMR cluster, which adds to operational overhead. Furthermore, it doesn't leverage Lake Formation for centralized governance.

- ⊗ Use AWS Glue Studio to ingest the data from the database to the S3 data lake. Attach an IAM policy to the QuickSight users to enforce column-level access control. Use Amazon S3 as the data source in QuickSight.

Using IAM policies directly on QuickSight users for column-level access control can become complex and difficult to manage, especially as the number of users and columns increases. It doesn't fully utilize Lake Formation's capabilities.

- ⊗ Use AWS Glue Elastic Views to create a materialized view for the database in Amazon S3. Create an S3 bucket policy to enforce column-level access control for the QuickSight users. Use Amazon S3 as the data source in QuickSight.

Managing column-level access control using S3 bucket policies becomes complex and error-prone. It's not a recommended approach for this scenario, and Lake Formation offers a better solution.

- ✔ Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight.

This solution leverages Lake Formation for both data ingestion and column-level access control, providing a centralized and manageable approach. Using Athena as the data source allows for querying the data lake with standard SQL and integrates seamlessly with Lake Formation's governance features. This minimizes operational overhead.

Overall Explanation:

The optimal solution is to use a Lake Formation blueprint to ingest data from Aurora MySQL into the S3 data lake. Lake Formation is then used to enforce column-level access control, streamlining permissions for QuickSight users. Using Athena as the data source

provides a SQL interface and minimizes the operational burden as it fully utilizes Lake Formation's features for centralized governance and access control.

[View Discussions](#)[Ask AI](#)

Question 30

Not Attempted

[Compute](#)

A transaction processing company runs weekly scripted batch jobs on Amazon EC2 instances within an Auto Scaling group. These jobs consistently utilize at least 60% CPU. The company needs to ensure capacity is provisioned 30 minutes before each job runs, but lacks the resources to analyze capacity trends for manual adjustments. The current method of manually modifying Auto Scaling group parameters is unsustainable. Which automated solution meets these requirements with the LEAST operational overhead?

- ☐ Create a dynamic scaling policy for the Auto Scaling group. Configure the policy to scale based on the CPU utilization metric. Set the target value for the metric to 60%.
Dynamic scaling reacts to current CPU utilization. It won't provision capacity in advance and won't solve the 30 minute requirement.
- ☒ Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.
Scheduled scaling allows pre-planning the capacity based on a fixed schedule. Although you have to guess, it is still much simpler than other scaling policies in this case.
- ☐ Create a predictive scaling policy for the Auto Scaling group. Configure the policy to scale based on forecast. Set the scaling metric to CPU utilization. Set the target value for the metric to 60%. In the policy, set the instances to pre-launch 30 minutes before the jobs run.
While predictive scaling can provision capacity in advance, the question states the company lacks resources to analyze capacity trends, which is needed for predictive scaling. It will be an overkill for the requirement.
- ☐ Create an Amazon EventBridge event to invoke an AWS Lambda function when the CPU utilization metric value for the Auto Scaling group reaches 60%. Configure the Lambda function to increase the Auto Scaling group's desired capacity and maximum capacity by 20%.

Restoring from S3 backups involves a manual process, including launching a new instance and restoring the database, thus increasing operational overhead compared to automated replication and failover solutions. While CRR replicates the backups, the restore process is not automatic.

Overall Explanation:

Migrating the MySQL database to an Amazon Aurora global database is the solution that best meets the requirements with the least operational overhead. Aurora is a managed database service, and its global database feature allows for replication and failover across multiple AWS Regions with minimal manual intervention. Hosting the primary DB cluster in the primary Region and the secondary DB cluster in the DR Region ensures that the data is available and up-to-date in both locations.

[View Discussions](#)[Ask AI](#)

Question 32

Not Attempted

[Application Integration](#)

A company has a Java application that uses Amazon Simple Queue Service (Amazon SQS) to parse messages. The application currently cannot parse messages that are larger than 256 KB in size. The company needs to enable the application to parse messages as large as 50 MB with minimal code changes. Which solution will meet these requirements most efficiently?

- ☒ Use the Amazon SQS Extended Client Library for Java to store messages larger than 256 KB in Amazon S3.

The Amazon SQS Extended Client Library for Java is designed to handle large messages by storing them in Amazon S3 and referencing them via SQS, allowing for messages up to 2 GB with minimal code changes.

- ☐ Use Amazon EventBridge to post large messages from the application instead of Amazon SQS.

While EventBridge is suitable for event routing, it doesn't inherently solve the large message size problem for the existing application designed to use SQS. It also requires significant code changes.

- ☐ Change the limit in Amazon SQS to handle messages that are larger than 256 KB. Amazon SQS has a hard limit of 256 KB for message sizes, which cannot be increased.

- ☐ Store messages larger than 256 KB in Amazon Elastic File System (Amazon EFS). Configure Amazon SQS to reference this location in the messages.

While possible, this solution requires managing file storage on EFS and ensuring the application can access it. The SQS Extended Client Library provides a more streamlined and managed approach using S3.

Overall Explanation:

The most efficient solution is to use the Amazon SQS Extended Client Library for Java. This library allows messages larger than 256 KB to be stored in Amazon S3 while SQS messages contain a reference to the object in S3. This method minimizes code changes, enabling the application to process messages up to 2 GB.

[View Discussions](#)[Ask AI](#)

Question 33

Not Attempted

Security, Identity, and Compliance



A company aims to secure its main web application by restricting content access and leveraging AWS authorization techniques. They seek a serverless architecture with an authentication solution for under 100 users. This solution must integrate seamlessly, deliver content globally, scale with user growth, and minimize login latency. Which solution fulfills these needs most cost-effectively?

- ☒ Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally.
Amazon Cognito offers serverless authentication for web apps. Lambda@Edge enables authorization with low latency. CloudFront delivers content globally, making this the most cost-effective solution.
- ☐ Use AWS Directory Service for Microsoft Active Directory for authentication. Use AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally.
Active Directory and ALB are more complex and expensive than Cognito and CloudFront, and do not inherently provide global content delivery with edge authorization.
- ☐ Use Amazon Cognito for authentication. Use AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally.
While Cognito provides authentication and S3 Transfer Acceleration enhances S3 transfer speeds, it lacks edge authorization which Lambda@Edge provides. Lambda running in a single region adds latency.

- ⊗ Use AWS Directory Service for Microsoft Active Directory for authentication. Use Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally.

Active Directory is an overkill for fewer than 100 users. Elastic Beanstalk is not a CDN and is more complex to manage than CloudFront.

Overall Explanation:

Amazon Cognito for authentication and Lambda@Edge for authorization, combined with Amazon CloudFront to serve the web application globally, provides a cost-effective solution with low-latency content delivery, global reach, and serverless architecture, fulfilling all requirements.

 View Discussions

 Ask AI

Question 34

Not Attempted

Storage



A company has an aging network-attached storage (NAS) array in its data center. The NAS array presents SMB shares and NFS shares to client workstations. The company does not want to purchase a new NAS array or renew the support contract. Some of the data is accessed frequently, but much of the data is inactive. A solutions architect needs to implement a solution that migrates the data to Amazon S3, uses S3 Lifecycle policies, and maintains the same look and feel for the client workstations. The solutions architect has identified AWS Storage Gateway as part of the solution. Which type of storage gateway should the solutions architect provision to meet these requirements?

- ⊗ Volume Gateway

Volume Gateway presents storage volumes to on-premises applications as iSCSI block devices. It's not suitable for presenting SMB or NFS shares and doesn't directly integrate with S3 Lifecycle policies.

- ⊗ Tape Gateway

Tape Gateway provides a virtual tape library (VTL) interface for backing up data to AWS. It's not relevant to migrating file shares and managing data lifecycle with S3.

- ⊗ Amazon FSx File Gateway

Amazon FSx File Gateway provides low-latency access to file systems in Amazon FSx for Windows File Server. While it supports SMB, it's designed for working with FSx file systems, not migrating from an existing NAS to S3.

✓ **Amazon S3 File Gateway**

Amazon S3 File Gateway provides a file interface (NFS and SMB) to objects stored in S3. It allows the company to migrate NAS data to S3 while maintaining the same look and feel for client workstations. S3 Lifecycle policies can be used to automate the movement of data to lower-cost storage tiers.

Overall Explanation:

Amazon S3 File Gateway is the correct choice because it presents SMB and NFS shares to the client workstations, allowing them to access data stored in Amazon S3. It also supports the use of S3 Lifecycle policies to transition inactive data to cheaper storage tiers. This solution addresses the company's need to migrate their NAS data to S3 while maintaining the same user experience and reducing storage costs.

 [View Discussions](#)

 [Ask AI](#)

Question 35

Not Attempted

[Compute](#)



A company runs an application on Amazon EC2 instances. The solutions architect has standardized the company on a particular instance family and various instance sizes based on current needs. The company aims to maximize cost savings for the application over the next 3 years. However, they anticipate needing to change the instance family and sizes within the next 6 months due to fluctuating application popularity and usage. Which solution will meet these requirements MOST cost-effectively?

✓ **Compute Savings Plan**

Compute Savings Plans offer flexibility and can reduce costs by up to 66%. They apply to EC2 instance usage regardless of instance family, size, AZ, Region, OS, or tenancy, and also apply to Fargate or Lambda usage.

✗ **EC2 Instance Savings Plan**

EC2 Instance Savings Plans provide savings in exchange for a commitment to a specific instance family. While they offer significant discounts, they do not allow changing the instance family.

✗ **Zonal Reserved Instances**

Zonal Reserved Instances offer cost savings for instances in a specific Availability Zone, but do not allow for the changing of instance families.

⊗ **Standard Reserved Instances**

Standard Reserved Instances don't provide the flexibility to easily change instance families. Convertible Reserved Instances would be more suitable if flexibility is needed.

Overall Explanation:

Compute Savings Plans are the MOST cost-effective solution because they provide the flexibility to change instance families and sizes as needed, while still offering significant cost savings. EC2 Instance Savings Plans lock you into a specific instance family, which is not suitable for the company's requirements. Zonal and Standard Reserved Instances do not provide the necessary flexibility to change instance families.

 [View Discussions](#)

 [Ask AI](#)

Question 36

Not Attempted

[Databases](#)



A company collects data from a large number of participants who use wearable devices. The company stores the data in an Amazon DynamoDB table and uses applications to analyze the data. The data workload is constant and predictable. The company wants to stay at or below its forecasted budget for DynamoDB. Which solution will meet these requirements MOST cost-effectively?

⊗ **Use provisioned mode and DynamoDB Standard-Infrequent Access (DynamoDB Standard-IA). Reserve capacity for the forecasted workload.**

DynamoDB Standard-IA is designed for tables where data is accessed infrequently. The scenario describes the data being actively analyzed, making Standard-IA unsuitable and less cost-effective.

✔ **Use provisioned mode. Specify the read capacity units (RCUs) and write capacity units (WCUs).**

Provisioned mode allows you to specify the RCUs and WCUs for your table, which is ideal for predictable workloads. By specifying the capacity based on the forecasted workload, you can optimize costs and avoid unnecessary expenses.

⊗ **Use on-demand mode. Set the read capacity units (RCUs) and write capacity units (WCUs) high enough to accommodate changes in the workload.**

On-demand mode does not allow you to set RCUs and WCUs. DynamoDB automatically manages capacity, which is more expensive than provisioned mode for predictable workloads.

- ⊗ Use on-demand mode. Specify the read capacity units (RCUs) and write capacity units (WCUs) with reserved capacity.

On-demand mode does not support specifying RCUs and WCUs, nor does it have a concept of reserved capacity.

Overall Explanation:

Since the data workload is constant and predictable, provisioned mode with specific RCUs and WCUs is the most cost-effective solution. On-demand mode is more expensive for predictable workloads, and DynamoDB Standard-IA is not appropriate for frequently accessed data. Therefore, the best option is to use provisioned mode and configure the RCUs and WCUs based on the forecasted workload.

[View Discussions](#)[Ask AI](#)

Question 37

Not Attempted

Databases



A company stores confidential data in an Amazon Aurora PostgreSQL database in the ap-southeast-3 Region. The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key. The company was recently acquired and must securely share a backup of the database with the acquiring company's AWS account in ap-southeast-3. What should a solutions architect do to meet these requirements?

- ⊗ Create a database snapshot. Copy the snapshot to a new unencrypted snapshot. Share the new snapshot with the acquiring company's AWS account.
Creating an unencrypted snapshot compromises the confidentiality of the data, which is not a secure solution.
- ✔ Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.
This is the most secure and recommended approach. By adding the acquiring company's account to the KMS key policy, they can decrypt the snapshot using the same key.
- ⊗ Create a database snapshot that uses a different AWS managed KMS key. Add the acquiring company's AWS account to the KMS key alias. Share the snapshot with the

acquiring company's AWS account.

Adding the acquiring company's AWS account to the KMS key alias doesn't directly control access to the encrypted data. KMS key aliases are simply alternative names for KMS keys and do not affect access control.

- ⊗ Create a database snapshot. Download the database snapshot. Upload the database snapshot to an Amazon S3 bucket. Update the S3 bucket policy to allow access from the acquiring company's AWS account.

Downloading and uploading the snapshot to an S3 bucket increases the risk of data leakage and is generally less secure than sharing the snapshot directly.

Overall Explanation:

The optimal solution involves creating a database snapshot and granting the acquiring company access to the KMS key used to encrypt the database. This is achieved by adding the acquiring company's AWS account ID to the KMS key policy, allowing them to decrypt the snapshot after it's shared. Options A and D introduce security vulnerabilities by creating unencrypted copies or transferring data insecurely. Option C is incorrect because adding the acquiring company's AWS account to the KMS key alias doesn't directly control access to the encrypted data. Access to encrypted data is governed by KMS key policies, which define who can use the key for encryption and decryption. Therefore, the most secure and correct approach is to add the acquiring company's account to the KMS key policy and then share the snapshot.

 View Discussions

 Ask AI

Question 38

Not Attempted

Security, Identity, and Compliance



A company is launching a public-facing web application in AWS, utilizing EC2 instances within a VPC behind an Elastic Load Balancer (ELB). A third-party service manages DNS. The solutions architect needs to recommend a solution for detecting and mitigating large-scale DDoS attacks. Which of the following options BEST meets these requirements?

- ⊗ Enable Amazon GuardDuty on the account.

Amazon GuardDuty focuses on detecting malicious activity and unauthorized behavior within AWS accounts, not specifically large-scale DDoS attacks.

⊗ Enable Amazon Inspector on the EC2 instances.

Amazon Inspector is a vulnerability assessment service and does not directly protect against DDoS attacks.

⊗ Enable AWS Shield and assign Amazon Route 53 to it.

While AWS Shield offers basic DDoS protection, the question mentions a third-party DNS service, making Route 53 integration irrelevant. Also, basic AWS Shield is not ideal for large-scale attacks.

✔ Enable AWS Shield Advanced and assign the ELB to it.

AWS Shield Advanced provides enhanced DDoS protection for resources like EC2 instances, ELBs, and more. Assigning the ELB ensures the application is protected against large-scale DDoS attacks.

Overall Explanation:

AWS Shield Advanced is designed to protect against large-scale DDoS attacks by providing expanded detection and mitigation capabilities specifically tailored for resources like ELBs, EC2 instances, and CloudFront distributions. GuardDuty and Inspector address different security concerns, while basic Shield with Route 53 doesn't offer the necessary protection and isn't compatible with the third-party DNS. Therefore, enabling AWS Shield Advanced and assigning the ELB to it is the most effective solution.

 View Discussions

 Ask AI

Question 39

Not Attempted

Databases



A company uses a 100 GB Amazon RDS for Microsoft SQL Server Single-AZ DB instance in the us-east-1 Region to store customer transactions. The company needs high availability and automatic recovery for the DB instance. The company must also run reports on the RDS database several times a year. The report process causes transactions to take longer than usual to post to the customers' accounts. The company needs a solution that will improve the performance of the report process. Which combination of steps will meet these requirements?

✔ Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment.

This will provide high availability and automatic recovery for the DB instance. If the primary DB instance fails, the standby DB instance will automatically become the primary DB instance.

- ⊗ Take a snapshot of the current DB instance. Restore the snapshot to a new RDS deployment in another Availability Zone.

This will not help improve the performance of the report process.

- ✔ Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica.

This will improve the performance of the report process by offloading the read traffic from the primary DB instance to the read replica.

- ⊗ Migrate the database to RDS Custom.

Migrating to RDS Custom does not directly address the need for high availability, automatic recovery and improving the report process performance.

- ⊗ Use RDS Proxy to limit reporting requests to the maintenance window.

RDS Proxy primarily helps with scalability and connection management, not performance of report generation. Limiting reporting requests to a maintenance window doesn't meet the availability requirements.

Overall Explanation:

Modifying the DB instance to a Multi-AZ deployment ensures high availability and automatic failover. Creating a read replica and directing report queries to it offloads the reporting workload from the primary instance, improving transaction posting times. Therefore options A and C are the correct choices.

 View Discussions

 Ask AI

Question 40

Not Attempted

Application Integration



A company is migrating its data management application to AWS and wants to adopt an event-driven architecture. They need a distributed, serverless solution to handle workflow steps with minimal operational overhead. Which solution best meets these requirements?

- ⊗ Build the workflow in AWS Glue and use it to invoke AWS Lambda functions for processing.

AWS Glue is primarily for ETL processes, not general workflow orchestration. While it can invoke Lambda, it's not the ideal choice for a distributed, event-driven architecture focused on minimizing operational overhead for general workflows.

- ☐ Build the workflow in AWS Step Functions, deploy the application on Amazon EC2 instances, and use Step Functions to invoke workflow steps on the EC2 instances.
This option defeats the purpose of serverless architecture by deploying the application on EC2 instances, increasing operational overhead.
- ☐ Build the workflow in Amazon EventBridge and use it to invoke AWS Lambda functions on a schedule for processing.
While EventBridge is event-driven, using a scheduled invocation doesn't fully utilize its capabilities for event-based workflows and may not be as responsive as needed. It is primarily targeted as an event router and not as a workflow engine.
- ☒ Build the workflow in AWS Step Functions, creating a state machine to invoke AWS Lambda functions for processing.
AWS Step Functions provides serverless workflow orchestration with state management, allowing for distributed application design and invocation of Lambda functions for each step, minimizing operational overhead. It is specifically designed for this type of use case.

Overall Explanation:

AWS Step Functions is the ideal solution because it's a serverless workflow orchestration service that enables building distributed applications by using state machines to coordinate AWS services like Lambda. This approach minimizes operational overhead while fully leveraging serverless concepts and delivering a truly event-driven architecture. Glue is ETL focused. EventBridge is an event router. Running on EC2 is not serverless.

[View Discussions](#)[Ask AI](#)

Question 41

Not Attempted

Networking



A company is designing the network for an online multi-player game. The game uses the UDP networking protocol and will be deployed in eight AWS Regions. The network architecture needs to minimize latency and packet loss to give end users a high-quality gaming experience. Which solution will meet these requirements?

- ☐ Set up a transit gateway in each Region. Create inter-Region peering attachments between each transit gateway.
Transit Gateway is designed to connect multiple VPCs but doesn't inherently optimize for low latency global UDP traffic. It mainly focuses on VPC to VPC connectivity, not end-user game

performance across regions.

✔ Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region.

AWS Global Accelerator supports UDP and TCP traffic and is designed to improve application availability and performance for global users. Using UDP listeners and endpoint groups in each region minimizes latency and packet loss for a high-quality gaming experience.

✘ Set up Amazon CloudFront with UDP turned on. Configure an origin in each Region.

CloudFront primarily handles HTTP and HTTPS content delivery. While it supports dynamic content, it's not designed for UDP-based real-time applications like online games, and doesn't natively support UDP.

✘ Set up a VPC peering mesh between each Region. Turn on UDP for each VPC.

While VPC peering allows connectivity between VPCs, creating a full mesh across eight regions becomes complex to manage. Furthermore, VPC peering doesn't inherently provide the low latency and packet loss reduction needed for a smooth gaming experience.

Overall Explanation:

AWS Global Accelerator is the best solution because it supports UDP traffic and is specifically designed to minimize latency and packet loss for applications accessed by global users. It provides static IP addresses as entry points and routes traffic to the optimal endpoint in the AWS Regions. Transit gateway is to connect multiple vpcs. CloudFront is not designed for UDP. VPC peering mesh is complex to manage across multiple regions.

 View Discussions

 Ask AI

Question 42

Not Attempted

Databases



A company hosts a three-tier web application on Amazon EC2 instances in a single Availability Zone. The web application uses a self-managed MySQL database that is hosted on an EC2 instance to store data in an Amazon Elastic Block Store (Amazon EBS) volume. The MySQL database currently uses a 1 TB Provisioned IOPS SSD (io2) EBS volume. The company expects traffic of 1,000 IOPS for both reads and writes at peak traffic. The company wants to minimize any disruptions, stabilize performance, and reduce costs while retaining the capacity for double the IOPS. The company wants to move the database tier to a fully managed solution that is highly available and fault tolerant. Which solution will meet these requirements MOST cost-effectively?

- ⊗ Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with an io2 Block Express EBS volume.

RDS does not support io2 Block Express. Also, using io2 is likely more expensive than necessary given the IOPS requirements can be met with gp2.

- ✔ Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume.

This option uses a fully managed RDS MySQL database with Multi-AZ for high availability and fault tolerance. The gp2 EBS volume is sufficient for the current IOPS requirement (2,000 IOPS) and offers cost-effectiveness.

- ⊗ Use Amazon S3 Intelligent-Tiering access tiers.

Amazon S3 is object storage and is not suitable for hosting a relational database.

- ⊗ Use two large EC2 instances to host the database in active-passive mode.

While this provides high availability, it requires manual management of failover and backups, and is not a fully managed solution. It is also less cost-effective than using RDS.

Overall Explanation:

The most cost-effective solution is to use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume. This satisfies the requirement for high availability, fault tolerance, and sufficient IOPS capacity while minimizing costs. gp2 EBS volume can provide sufficient IOPS within the cost constraint, and RDS manages the database administration overhead. The other options are either not cost-effective or do not meet the requirements.

 View Discussions

 Ask AI

Question 43

Not Attempted

Databases



A company hosts a serverless application on AWS using Amazon API Gateway, AWS Lambda, and an Amazon RDS for PostgreSQL database. They are experiencing increased application errors due to database connection timeouts during peak traffic. They need a solution to minimize these failures with minimal code changes. What should a solutions architect do?

- ⊗ Reduce the Lambda concurrency rate.

Reducing Lambda concurrency might limit the number of requests, but it does not address the underlying issue of connection exhaustion and doesn't directly help with connection timeouts.

✔ Enable RDS Proxy on the RDS DB instance.

RDS Proxy is a fully managed, highly available database proxy that helps manage database connections, reducing the load on the database and minimizing connection timeouts. It also requires minimal code changes.

✘ Resize the RDS DB instance class to accept more connections.

While increasing the instance size might allow for more connections, it doesn't solve the root cause of inefficient connection management and could lead to increased costs without necessarily resolving the timeout issues. Also, more connections may lead to worse performance.

✘ Migrate the database to Amazon DynamoDB with on-demand scaling.

Migrating to DynamoDB involves significant code changes and a database schema redesign. It might be a viable option in some scenarios, but it's not the best solution when the goal is to minimize code changes and address the immediate problem of connection timeouts.

Overall Explanation:

RDS Proxy is the best solution because it pools and shares database connections, reducing the number of open connections to the database and minimizing connection timeouts. It requires minimal code changes and directly addresses the problem described in the scenario.

 View Discussions

 Ask AI

Question 44

Not Attempted

Compute



A company is migrating an old application to AWS. The application runs a CPU-intensive batch job every hour, taking 15 minutes on average on an on-premises server with 64 vCPUs and 512 GiB of memory. The company needs to run this job on AWS within 15 minutes with the LEAST operational overhead. Which solution should they use?

✘ Use AWS Lambda with functional scaling.

AWS Lambda has limits on execution time (maximum 15 minutes which is equal to batch time but on average), memory, and vCPU cores. The stated memory and vCPU requirements exceed Lambda's capabilities. Also, Lambda might not be ideal for long-running, CPU-intensive batch jobs, as it's designed for event-driven functions.

⊗ Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate.

AWS Fargate offers containerized application orchestration and scalable infrastructure. However, it may require more operational overhead to configure and manage the environment compared to AWS Batch.

⊗ Use Amazon Lightsail with AWS Auto Scaling.

Amazon Lightsail has limits on vCPUs and memory, making it unsuitable for the specified resource requirements. It is difficult to get the required vCPU's and memory.

✔ Use AWS Batch on Amazon EC2.

AWS Batch is a fully managed service specifically designed for running batch jobs. It automatically provisions and manages the necessary infrastructure, allowing for the use of Amazon EC2 instances with the required vCPU and memory resources. It also handles job scheduling, instance management, and scaling automatically, minimizing operational overhead.

Overall Explanation:

AWS Batch is the most suitable solution because it's a fully-managed service that simplifies running batch jobs on AWS. It can easily scale compute resources (EC2 instances) to match the on-premises server's vCPU and memory, handle job scheduling and instance management automatically, and offers lower operational overhead compared to ECS/Fargate or Lightsail. Lambda's resource limits make it unsuitable for this specific workload. Therefore, using AWS Batch on Amazon EC2 provides the best combination of performance and operational simplicity.

 View Discussions

 Ask AI

Question 45

Not Attempted

Storage



A company stores its data objects in Amazon S3 Standard storage. A solutions architect has determined that 75% of the data is rarely accessed after 30 days. The company requires all data to remain immediately accessible with the same high availability and resiliency, but wants to minimize storage costs. Which storage solution will meet these requirements?

⊗ Move the data objects to S3 Glacier Deep Archive after 30 days.

S3 Glacier Deep Archive is intended for archiving data that is rarely accessed and can tolerate retrieval times measured in hours, which does not meet the immediate accessibility requirement.

- ✔ Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.
S3 Standard-IA is designed for infrequently accessed data, providing lower storage costs than S3 Standard while still offering immediate accessibility, high availability, and high resiliency.
- ✘ Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.
S3 One Zone-IA stores data in a single Availability Zone, which does not provide the same high availability and resiliency as S3 Standard or S3 Standard-IA.
- ✘ Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately.
Moving the data to S3 One Zone-IA immediately, before the 30-day mark, doesn't address the initial requirement of storing the data in S3 Standard for the first 30 days. Also, S3 One Zone-IA does not provide the same high availability and resiliency.

Overall Explanation:

S3 Standard-IA is the most appropriate option as it balances the need for cost optimization with the requirement for immediate accessibility, high availability, and resiliency. S3 Glacier Deep Archive has long retrieval times, and S3 One Zone-IA lacks the required high availability.

[View Discussions](#)[Ask AI](#)

Question 46

Not Attempted

Storage



A gaming company is migrating its public scoreboard application from an on-premises data center to the AWS Cloud. The company uses Amazon EC2 Windows Server instances behind an Application Load Balancer to host the dynamic application. The application requires a highly available storage solution for both static files and dynamic server-side code. Which combination of actions should a solutions architect take to meet these requirements?

- ✔ Store the static files on Amazon S3. Use Amazon CloudFront to cache objects at the edge.
Storing static files in S3 provides durability and scalability. CloudFront caches these files at edge locations, reducing latency for users accessing the scoreboard.

- ⊗ Store the static files on Amazon S3. Use Amazon ElastiCache to cache objects at the edge.
While ElastiCache is a caching service, it is not designed to cache content at the edge like a CDN (Content Delivery Network). CloudFront is more suitable for static content delivery to users globally.
- ⊗ Store the server-side code on Amazon Elastic File System (Amazon EFS). Mount the EFS volume on each EC2 instance to share the files.
Amazon EFS is primarily designed for Linux-based workloads and is not directly compatible with Windows Server instances.
- ✔ Store the server-side code on Amazon FSx for Windows File Server. Mount the FSx for Windows File Server volume on each EC2 instance to share the files.
Amazon FSx for Windows File Server provides a fully managed, highly available Windows native file system that can be accessed from the Windows EC2 instances to share server-side code. It is designed for high availability and scales up to 10s of GBPS throughput, making it suitable for this purpose.
- ⊗ Store the server-side code on a General Purpose SSD (gp2) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on each EC2 instance to share the files.
EBS volumes are block storage devices and are not designed for sharing files between multiple EC2 instances. EBS volumes can only be attached to a single EC2 instance at a time.

Overall Explanation:

The correct combination is to store static files on Amazon S3 with CloudFront for caching at the edge and to store server-side code on Amazon FSx for Windows File Server, mounting the FSx volume on each EC2 instance. S3/CloudFront provides highly available and scalable storage for static assets with low latency access, and FSx for Windows File Server allows the Windows EC2 instances to share the dynamic server-side code with high performance and availability. EFS is not compatible with Windows, and EBS volumes cannot be shared among EC2 instances.

[View Discussions](#)[Ask AI](#)

Question 47

Not Attempted

[Compute](#)

A social media company hosts its application on Amazon EC2 instances behind an Application Load Balancer (ALB). The ALB serves as the origin for an Amazon CloudFront distribution. The

application stores over a billion images in an Amazon S3 bucket and processes thousands of images each second. The company requires a solution to dynamically resize these images and deliver appropriate formats to various clients with minimal operational overhead. Which solution is the MOST suitable?

- ☐ Install an external image management library on an EC2 instance. Utilize the library to process the images.
- ☐ Create a CloudFront origin request policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

Installing and managing image processing on EC2 instances introduces significant operational overhead, including scaling, patching, and ensuring high availability.

CloudFront origin request policies cannot directly resize images. They primarily manipulate request headers or query strings passed to the origin.

- ☒ Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors that serve the images.

Lambda@Edge enables image processing at the edge, closer to the user, reducing latency. Using an external image management library within Lambda@Edge allows for dynamic resizing and format conversion based on client capabilities with minimal operational overhead, as it is a serverless solution.

- ☐ Create a CloudFront response headers policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

CloudFront response headers policies are used to add or modify HTTP response headers and cannot be used to resize images or dynamically modify content based on the User-Agent header.

Overall Explanation:

Lambda@Edge, combined with an image management library, provides a serverless solution for dynamic image resizing and format conversion at the edge. This minimizes latency and operational overhead, making it the most efficient approach for meeting the given requirements. CloudFront origin/response header policies are not designed to perform image manipulation. EC2 instances introduce management overhead.

 View Discussions

 Ask AI

Question 48

Not Attempted

Security, Identity, and Compliance



A hospital needs to store patient records in an Amazon S3 bucket. The hospital's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest. The compliance team must administer the encryption key for data at rest. Which solution will meet these requirements?

- ⊗ Create a public SSL/TLS certificate in AWS Certificate Manager (ACM). Associate the certificate with Amazon S3. Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

ACM certificates are not directly associated with S3 buckets. While SSE-KMS is correct for encryption at rest with compliance team key management, the ACM part is incorrect.

- ⊗ Use the `aws:SecureTransport` condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with S3 managed encryption keys (SSE-S3). Assign the compliance team to manage the SSE-S3 keys.

SSE-S3 uses keys managed by AWS, not the compliance team. The `aws:SecureTransport` condition is correct for encryption in transit, but the key management aspect is incorrect.

- ✔ Use the `aws:SecureTransport` condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

This solution correctly addresses both encryption in transit (`aws:SecureTransport`) and encryption at rest with compliance team-managed keys (SSE-KMS).

- ⊗ Use the `aws:SecureTransport` condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Use Amazon Macie to protect the sensitive data that is stored in Amazon S3. Assign the compliance team to manage Macie.

Amazon Macie is a data security and data privacy service that uses machine learning to discover, classify, and protect sensitive data stored in S3. Macie itself does not encrypt the data in the way the question is asking. While Macie can identify sensitive data, it doesn't provide the encryption management required. It also encrypts its own findings using AWS managed keys, removing key admin responsibility from the compliance team.

Overall Explanation:

Option C is the correct answer. Using the `aws:SecureTransport` condition ensures that all data is encrypted in transit using HTTPS. Configuring server-side encryption with AWS KMS keys (SSE-KMS) allows the compliance team to administer the encryption keys for the data at rest, fulfilling the requirement. Options A, B, and D do not fully meet the requirements as they either incorrectly implement the encryption in transit, the key management, or attempt to use the wrong AWS service.

[View Discussions](#)[Ask AI](#)

Question 49

Not Attempted

Security, Identity, and Compliance



A company is developing an application on AWS that requires data to be stored in Amazon S3 buckets across two AWS Regions for redundancy. The company mandates that all data at rest in these S3 buckets be encrypted using an AWS Key Management Service (AWS KMS) customer managed key. Both S3 buckets must use the same KMS key for encryption and decryption, and the key material must be available in both Regions. The solution should minimize operational overhead. Which approach best meets these requirements?

- ☐ Create an S3 bucket in each Region. Configure the S3 buckets to use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Configure cross-region replication between the S3 buckets.

This option does not meet the requirement of using a customer managed key; it uses S3 managed keys.

- ☐ Create a customer managed multi-Region KMS key. Create an S3 bucket in each Region. Configure cross-region replication between the S3 buckets. Configure the application to use the KMS key with client-side encryption.

Client-side encryption increases operational overhead as the application needs to handle encryption and decryption. Although Multi-Region KMS is possible, the client-side encryption is not optimal.

- ☐ Create a customer managed KMS key and an S3 bucket in each Region. Configure the S3 buckets to use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Configure cross-region replication between the S3 buckets.

This option uses SSE-S3, which means that S3 manages the encryption keys, not the customer.

- ☒ Create a customer managed KMS key and an S3 bucket in each Region. Configure the S3 buckets to use server-side encryption with AWS KMS keys (SSE-KMS). Configure

cross-region replication between the S3 buckets.

This option meets all requirements: data is encrypted with a customer managed KMS key, encryption is done server-side, and the key material is available to both regions. Replication ensures that the data is in both regions.

Overall Explanation:

The optimal solution is to use server-side encryption with AWS KMS keys (SSE-KMS) using a customer managed key. This minimizes operational overhead as S3 handles the encryption and decryption. Creating S3 buckets in each region and enabling cross-region replication fulfills the availability requirement.

[View Discussions](#)[Ask AI](#)

Question 50

Not Attempted

Networking



A company uses Amazon API Gateway to run a private gateway with two REST APIs in the same VPC. The `BuyStock` RESTful web service calls the `CheckFunds` RESTful web service to ensure that enough funds are available before a stock can be purchased. The company has noticed in the VPC flow logs that the `BuyStock` RESTful web service calls the `CheckFunds` RESTful web service over the internet instead of through the VPC. A solutions architect must implement a solution so that the APIs communicate through the VPC. Which solution will meet these requirements with the FEWEST changes to the code?

☐ Add an X-API-Key header in the HTTP header for authorization.

Adding an X-API-Key header for authorization doesn't address the issue of ensuring that the APIs communicate through the VPC. It's more related to authentication and authorization mechanisms.

☒ Use an interface endpoint.

An interface endpoint is a horizontally scaled, redundant VPC endpoint that provides private connectivity to a service. It is an elastic network interface with a private IP address that serves as an entry point for traffic destined to the AWS service. Interface endpoints are used to connect VPCs with AWS services, ensuring communication within the VPC without traversing the public internet.

☐ Use a gateway endpoint.

Gateway endpoints only support Amazon S3 and DynamoDB, not generic REST APIs.

- ⊗ Add an Amazon Simple Queue Service (Amazon SQS) queue between the two REST APIs.

Adding an SQS queue introduces asynchronous communication and would require significant code changes to both APIs.

Overall Explanation:

The best solution is to use an interface endpoint. An interface endpoint allows communication between resources in your VPC and services without traversing the public internet. You can create an interface endpoint for API Gateway in your VPC. This enables the communication between the BuyStock and CheckFunds RESTful web services within the VPC, and it doesn't require significant changes to the code. Gateway endpoints only support S3 and DynamoDB, and using SQS would necessitate larger code modifications.

 View Discussions

 Ask AI

Question 51

Not Attempted

Databases



A company hosts a multiplayer gaming application on AWS. The company wants the application to read data with sub-millisecond latency and run one-time queries on historical data. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Use Amazon RDS for data that is frequently accessed. Run a periodic custom script to export the data to an Amazon S3 bucket.

Using RDS and a custom script introduces operational overhead for managing the script and exporting data. It also doesn't inherently provide sub-millisecond latency.

- ⊗ Store the data directly in an Amazon S3 bucket. Implement an S3 Lifecycle policy to move older data to S3 Glacier Deep Archive for long-term storage. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

Storing all data in S3 doesn't meet the sub-millisecond latency requirement for frequently accessed data.

- ✓ Use Amazon DynamoDB with DynamoDB Accelerator (DAX) for data that is frequently accessed. Export the data to an Amazon S3 bucket by using DynamoDB table export. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

This option provides the best balance of low latency for frequently accessed data using DynamoDB with DAX, and cost-effective querying of historical data in S3 using Athena. DynamoDB export to S3

simplifies the export process, minimizing operational overhead.

- ⊗ Use Amazon DynamoDB for data that is frequently accessed. Turn on streaming to Amazon Kinesis Data Streams. Use Amazon Kinesis Data Firehose to read the data from Kinesis Data Streams. Store the records in an Amazon S3 bucket.

Using Kinesis Data Streams and Firehose adds unnecessary complexity and latency for this use case, and increases operational overhead.

Overall Explanation:

DynamoDB with DAX provides sub-millisecond latency for frequently accessed data.

DynamoDB table export to S3, combined with Athena, allows for efficient one-time queries on historical data with minimal operational overhead. DAX delivers up to a 10 times performance improvement—from milliseconds to microseconds. Using DynamoDB export to S3, you can export data from an Amazon DynamoDB table to an Amazon S3 bucket for analytics and complex queries.

 View Discussions

 Ask AI

Question 52

Not Attempted

Application Integration



A company uses a payment processing system that requires messages for a particular payment ID to be received and processed in the same order that they were sent. Otherwise, the payments might be processed incorrectly. Which actions should a solutions architect take to meet this requirement? (Choose two.)

- ⊗ Write the messages to an Amazon DynamoDB table with the payment ID as the partition key.

DynamoDB does not guarantee message order within a partition, so this option is incorrect.

- ✓ Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key.

Kinesis Data Streams maintains the order of records within a shard. Using the payment ID as the partition key ensures all messages for that ID are sent to the same shard, preserving order.

- ☐ Write the messages to an Amazon ElastiCache for Memcached cluster with the payment ID as the key.
ElastiCache is a caching service and does not guarantee message ordering.
- ☐ Write the messages to an Amazon Simple Queue Service (Amazon SQS) queue. Set the message attribute to use the payment ID.
Standard SQS queues do not guarantee message order.
- ☒ Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID.
SQS FIFO queues guarantee that messages are received in the exact order they are sent. Using the payment ID as the message group ensures all messages for a payment ID are received sequentially.

Overall Explanation:

To ensure messages for a payment ID are processed in order, you can either use Amazon Kinesis Data Streams or Amazon SQS FIFO queues. Kinesis maintains order within a shard using the partition key. SQS FIFO queues guarantee message order within a message group. DynamoDB, ElastiCache and standard SQS do not guarantee message ordering and are therefore not suitable.

[View Discussions](#)[Ask AI](#)

Question 53

Not Attempted

Application Integration



A company is developing a real-time multiplayer game on AWS. The game requires a system that can reliably deliver in-game events to multiple backend services (leaderboard, matchmaking, and player authentication) in the order they occurred. The company needs an event-driven architecture that guarantees event ordering and concurrent delivery to these separate services. Which AWS service configuration best meets these requirements?

- ☐ Amazon EventBridge event bus with content-based filtering rules targeting individual services.
EventBridge doesn't guarantee event ordering across multiple targets.
- ☒ Amazon Simple Notification Service (Amazon SNS) FIFO topics with separate subscriptions for each backend service.

SNS FIFO topics ensure ordered message delivery and support fan-out to multiple subscribers.

- ⊗ Amazon Simple Notification Service (Amazon SNS) standard topics with separate subscriptions for each backend service.

SNS standard topics do not guarantee message order.

- ⊗ Amazon Simple Queue Service (Amazon SQS) FIFO queues for each backend service, with the game server publishing directly to each queue.

While SQS FIFO guarantees order, using it directly for multiple destinations is less efficient than SNS for fan-out scenarios.

Overall Explanation:

Amazon SNS FIFO topics provide the necessary features: guaranteed ordering and the ability to publish messages to multiple subscribers (in this case, the leaderboard, matchmaking, and authentication services) concurrently. The other options either don't guarantee ordering (EventBridge, SNS standard) or are not optimized for a pub/sub fan-out pattern (SQS FIFO). Using SNS FIFO topics with separate subscriptions for each service allows each service to receive events in the correct order without affecting the others.

 View Discussions

 Ask AI

Question 54

Not Attempted

Security, Identity, and Compliance



A hospital is designing a new application that gathers symptoms from patients. The hospital has decided to use Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS) in the architecture. A solutions architect is reviewing the infrastructure design. Data must be encrypted at rest and in transit. Only authorized personnel of the hospital should be able to access the data. Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

- ⊗ Turn on server-side encryption on the SQS components. Update the default key policy to restrict key usage to a set of authorized principals.

Incorrect. You don't update the *default* key policy; you create or modify a key policy.

- ✓ Turn on server-side encryption on the SNS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict

key usage to a set of authorized principals.

Correct. Using a KMS customer managed key allows for control of the encryption key and its policy, restricting access to authorized personnel.

- ⊗ Turn on encryption on the SNS components. Update the default key policy to restrict key usage to a set of authorized principals. Set a condition in the topic policy to allow only encrypted connections over TLS.

Incorrect. You cannot update the default key policy. Also, while TLS encryption is required for SNS with SSE enabled, the first part of the sentence uses generic phrasing and not specific enough for this scenario.

- ✓ Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS.

Correct. Using KMS with a customer managed key allows for control of the encryption key and its policy, restricting access. Setting a TLS policy ensures encryption in transit.

- ⊗ Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply an IAM policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS.

Incorrect. IAM policies cannot specify the principal within the policy in this context. Key policies are required to add permissions for each queue producer and consumer. Also, the question requests to restrict access to *authorized personnel*, and IAM cannot be used for this.

Overall Explanation:

To meet the requirements, encryption at rest is achieved by enabling server-side encryption (SSE) on both SQS and SNS using KMS customer managed keys. A key policy restricts key usage to authorized personnel. Encryption in transit is enforced using TLS. Options B and D correctly address these requirements.

 View Discussions

 Ask AI

Question 55

Not Attempted

Databases



A company runs a web application backed by Amazon RDS. A database administrator accidentally corrupted data in a database table. The company needs to restore the database to

its state from 5 minutes before the change within the last 30 days to recover from such incidents. Which feature should a solutions architect include in the design to meet this requirement?

☐ Read replicas

Read replicas are used for offloading read traffic and improving read performance, not for point-in-time recovery.

☐ Manual snapshots

While manual snapshots can be used for backups, they don't provide the granularity needed for restoring to a specific point in time 5 minutes before a change within the last 30 days without a complex manual process.

☒ Automated backups

Automated backups provide point-in-time recovery within the retention period (up to 35 days). They allow restoring the database to its state from any point within the retention period, including 5 minutes before a change.

☐ Multi-AZ deployments

Multi-AZ deployments provide high availability and failover capabilities but do not, on their own, offer the ability to restore the database to a specific point in time before a data corruption event.

Overall Explanation:

Automated backups enable point-in-time recovery (PITR) for your DB instance down to a specific second within the retention period, which can be up to 35 days. By setting the retention period to 30 days, the company can restore the database to its state from up to 5 minutes before any change within the last 30 days. The recovery process creates a new Amazon RDS instance with a new endpoint.

 [View Discussions](#)

 [Ask AI](#)

Question 56

Not Attempted

[Security, Identity, and Compliance](#)



A company's web application consists of an Amazon API Gateway API in front of an AWS Lambda function and an Amazon DynamoDB database. The Lambda function handles the business logic, and the DynamoDB table hosts the data. The application uses Amazon Cognito user pools to identify individual users. A solutions architect needs to update the application so

that only users who have a subscription can access premium content. Which solution will meet this requirement with the LEAST operational overhead?

- ☐ Enable API caching and throttling on the API Gateway API.
API caching and throttling optimize and control API usage but do not limit access based on subscriptions.
- ☐ Set up AWS WAF on the API Gateway API. Create a rule to filter users who have a subscription.
Using AWS WAF could work, but requires ongoing management of WAF rules and increases operational overhead.
- ☐ Apply fine-grained IAM permissions to the premium content in the DynamoDB table.
This is a viable approach. However, managing fine-grained IAM permissions for DynamoDB access based on subscriptions adds operational overhead.
- ☒ Implement API usage plans and API keys to limit the access of users who do not have a subscription.
Implementing API usage plans and API keys is a straightforward way to restrict access to specific users or groups based on subscriptions with minimal operational overhead. It allows control at the API level without extensive architectural changes.

Overall Explanation:

Implementing API usage plans and API keys allows you to control access at the API level and doesn't require extensive changes to your existing architecture. This solution provides a clear and manageable way to enforce access restrictions without complicating other parts of the application. Options A, B, and C either don't address the core requirement or introduce more operational overhead than option D.

[View Discussions](#)[Ask AI](#)

Question 57

Not Attempted

Networking



A company is using Amazon Route 53 latency-based routing to route requests to its UDP-based application for users around the world. The application is hosted on redundant servers in the company's on-premises data centers in the United States, Asia, and Europe. The company's compliance requirements state that the application must be hosted on premises. The company

wants to improve the performance and availability of the application. What should a solutions architect do to meet these requirements?

- ✔ Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and register the NLBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS.

This is the correct answer. NLBs support UDP traffic, which is required by the application. AWS Global Accelerator uses Anycast IP addresses and its global network to route users to the optimal endpoint, improving performance and availability. Registering NLBs as Global Accelerator endpoints provides improved availability and DDoS protection.

- ✘ Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and register the ALBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS.

This is incorrect because ALBs do not support UDP traffic.

- ✘ Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three NLBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

This is incorrect because CloudFront does not support UDP traffic and using CloudFront means the application isn't strictly hosted on-premise.

- ✘ Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three ALBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

This is incorrect because ALBs and CloudFront do not support UDP traffic.

Overall Explanation:

The correct solution is to use Network Load Balancers (NLBs) with AWS Global Accelerator. NLBs are required for UDP traffic. Global Accelerator improves performance and availability by routing traffic to the optimal endpoint using Anycast IP addresses. Options using ALB or CloudFront are incorrect because they don't support UDP. Additionally, Global Accelerator is the better choice over latency-based routing directly through Route53 for performance improvements.

[View Discussions](#)[Ask AI](#)

Question 58

Not Attempted

Security, Identity, and Compliance



A solutions architect wants all new users to have specific complexity requirements and mandatory rotation periods for IAM user passwords. What should the solutions architect do to accomplish this?

- ☒ Set an overall password policy for the entire AWS account.
Setting an account-level password policy ensures that all IAM users, including new ones, adhere to the specified complexity and rotation rules. This is the recommended and most efficient approach.
- ☐ Set a password policy for each IAM user in the AWS account.
IAM does not allow password policies to be set individually for each user. Password policies are applied at the account level.
- ☐ Use third-party vendor software to set password requirements.
While third-party solutions exist, using native AWS IAM password policies is the simplest and most cost-effective approach for basic password management.
- ☐ Attach an Amazon CloudWatch rule to the Create_newuser event to set the password with the appropriate requirements.
CloudWatch Events can trigger actions based on AWS events, it is not the appropriate way to enforce password policies. Password policies are set at the account level within IAM.

Overall Explanation:

The most efficient way to enforce password complexity requirements and mandatory rotation periods for all new IAM users is to set an overall password policy for the entire AWS account. This ensures that all users, present and future, are subject to the same password rules. Setting an account-level policy is the direct approach offered by AWS for managing user passwords. Other approaches are either inefficient or not possible using native AWS services.

[View Discussions](#)[Ask AI](#)

Question 59

Not Attempted

Compute



A company has migrated an application to Amazon EC2 Linux instances. One of these EC2 instances runs several 1-hour tasks on a schedule. These tasks were written by different teams and have no common programming language. The company is concerned about performance and scalability while these tasks run on a single instance. A solutions architect needs to implement a solution to resolve these concerns. Which solution will meet these requirements with the LEAST operational overhead?

- ☒ Use AWS Batch to run the tasks as jobs. Schedule the jobs by using Amazon EventBridge (Amazon CloudWatch Events).

AWS Batch is designed for running batch computing workloads and can handle tasks written in different languages. It manages the underlying compute resources, reducing operational overhead. It is appropriate for scheduled jobs.

- ☐ Convert the EC2 instance to a container. Use AWS App Runner to create the container on demand to run the tasks as jobs.

Converting the EC2 instance to a container and using AWS App Runner introduces additional operational overhead compared to AWS Batch. AWS Batch directly manages the underlying compute resources for the tasks.

- ☐ Copy the tasks into AWS Lambda functions. Schedule the Lambda functions by using Amazon EventBridge (Amazon CloudWatch Events).

AWS Lambda has a maximum execution time of 15 minutes, which is insufficient for the 1-hour tasks. Therefore, this is not a viable solution.

- ☐ Create an Amazon Machine Image (AMI) of the EC2 instance that runs the tasks. Create an Auto Scaling group with the AMI to run multiple copies of the instance.

Creating an AMI and using an Auto Scaling group adds unnecessary operational overhead for scheduled tasks. Auto Scaling is more suitable for responding to changes in demand, rather than running scheduled jobs. Also, this does not address the performance concerns if the tasks run sequentially on each instance.

Overall Explanation:

AWS Batch provides a fully managed environment for running batch computing workloads. It supports diverse programming languages and dynamically provisions resources, minimizing operational overhead. Since the tasks run for an hour, Lambda is not an option. Auto Scaling isn't ideal for scheduled tasks, making Batch the most suitable answer.

[View Discussions](#)[Ask AI](#)

Question 60

Not Attempted

Security, Identity, and Compliance



A company recently launched a variety of new workloads on Amazon EC2 instances in its AWS account. The company needs to create a strategy to access and administer the instances remotely and securely. The company needs to implement a repeatable process that works with native AWS services and follows the AWS Well-Architected Framework. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Use the EC2 serial console to directly access the terminal interface of each instance for administration.

The EC2 serial console provides direct access but is cumbersome to manage, especially for multiple instances and is not ideal for remote administration at scale.

- ☒ Attach the appropriate IAM role to each existing instance and new instance. Use AWS Systems Manager Session Manager to establish a remote SSH session.

This solution uses IAM roles for secure access and Session Manager for remote access without managing SSH keys or bastion hosts, resulting in the least operational overhead. It leverages native AWS services and is easily repeatable.

- ☐ Create an administrative SSH key pair. Load the public key into each EC2 instance. Deploy a bastion host in a public subnet to provide a tunnel for administration of each instance.

A bastion host adds operational overhead by requiring management, monitoring, and securing an additional infrastructure component. SSH key management also increases operational burden.

- ☐ Establish an AWS Site-to-Site VPN connection. Instruct administrators to use their local on-premises machines to connect directly to the instances by using SSH keys across the VPN tunnel.

Establishing a VPN connection is complex and unnecessary for simple remote administration. Connecting from on-premises machines adds complexity and potential security risks.

Overall Explanation:

The optimal solution is to utilize AWS Systems Manager Session Manager in conjunction with IAM roles. Session Manager provides secure, auditable, and repeatable remote access

without the need for SSH keys or bastion hosts, minimizing operational overhead. IAM roles ensure secure access control and align with the AWS Well-Architected Framework.

[View Discussions](#)[Ask AI](#)

Question 61

Not Attempted

Networking



A company hosts a public three-tier web application within an Amazon VPC. The application is deployed on Amazon EC2 instances distributed across multiple Availability Zones. EC2 instances residing in private subnets require outbound internet access to communicate with a license server. The company seeks a managed solution that minimizes operational overhead. Which of the following solutions best meets these requirements?

- ☐ Provision a NAT instance in a public subnet. Configure each private subnet's route table to use the NAT instance as the default route.

Using a NAT instance requires manual configuration and ongoing maintenance, contradicting the requirement for a managed solution with minimal operational overhead.

- ☐ Provision a NAT instance in a private subnet. Configure each private subnet's route table to use the NAT instance as the default route.

A NAT instance in a private subnet cannot route traffic to the internet, and also doesn't meet the requirement for a managed solution with minimal operational overhead.

- ☒ Provision a NAT gateway in a public subnet. Configure each private subnet's route table to use the NAT gateway as the default route.

A NAT Gateway in a public subnet provides a managed, highly available, and scalable solution for instances in private subnets to access the internet. This option minimizes operational maintenance.

- ☐ Provision a NAT gateway in a private subnet. Configure each private subnet's route table to use the NAT gateway as the default route.

NAT Gateways must be placed in public subnets to function correctly. They cannot be deployed within private subnets.

Overall Explanation:

The most suitable solution is to provision a NAT gateway within a public subnet and configure the private subnet route tables to route internet-bound traffic through the NAT gateway. NAT gateways are managed services, thus minimizing operational overhead.

Furthermore, they provide high availability and scalability. NAT gateways must reside in a public subnet to facilitate internet connectivity.

[View Discussions](#)[Ask AI](#)

Question 62

Not Attempted

Security, Identity, and Compliance



A company needs to create an Amazon Elastic Kubernetes Service (Amazon EKS) cluster to host a digital media streaming application. The EKS cluster will use a managed node group that is backed by Amazon Elastic Block Store (Amazon EBS) volumes for storage. The company must encrypt all data at rest by using a customer managed key that is stored in AWS Key Management Service (AWS KMS). Which combination of actions will meet this requirement with the LEAST operational overhead?

- ☐ Use a Kubernetes plugin that uses the customer managed key to perform data encryption.
Using a Kubernetes plugin introduces additional complexity and overhead compared to native AWS solutions.
- ☐ After creation of the EKS cluster, locate the EBS volumes. Enable encryption by using the customer managed key.
While this works, manually enabling encryption on each EBS volume after creation adds significant operational overhead, especially as the cluster scales.
- ☒ Enable EBS encryption by default in the AWS Region where the EKS cluster will be created. Select the customer managed key as the default key.
Enabling EBS encryption by default encrypts all new EBS volumes created in the region with the specified KMS key. This is a simple, region-wide setting that minimizes operational overhead and satisfies the 'encrypt all data at rest' requirement.
- ☒ Create the EKS cluster. Create an IAM role that has a policy that grants permission to the customer managed key. Associate the role with the EKS cluster.
The EKS cluster needs permissions to use the KMS key for encryption. Creating an IAM role with the necessary permissions and associating it with the cluster ensures that the cluster can access the KMS key.
- ☐ Store the customer managed key as a Kubernetes secret in the EKS cluster. Use the customer managed key to encrypt the EBS volumes.

Storing KMS keys as Kubernetes secrets is a security risk. KMS keys should be managed by AWS KMS.

Overall Explanation:

Options C and D provide the least operational overhead while meeting the requirement. Option C enables EBS encryption by default for all newly created EBS volumes in the region, ensuring that all EBS volumes associated with the EKS cluster are encrypted with the customer-managed KMS key without requiring manual intervention for each volume. Option D configures the EKS cluster with the necessary IAM permissions to access the KMS key, allowing the cluster to use it for encryption and decryption operations.

[View Discussions](#)[Ask AI](#)

Question 63

Not Attempted

[Databases](#)

A company is migrating an Oracle database to AWS. The database stores millions of high-resolution Geographic Information System (GIS) images, each identified by a unique geographic code. During natural disasters, tens of thousands of images are updated within minutes. Each geographic code corresponds to a single image record. The company needs a highly available and scalable solution for these peak update events that is also cost-effective. Which solution BEST meets these requirements?

- ☐ Store images and geographic codes in an Oracle database running on an Amazon RDS Multi-AZ DB instance.

Using Oracle RDS, even with Multi-AZ, may not be the most cost-effective solution for this specific use case, especially considering the large volume of image updates. RDS, being a relational database, might not be the ideal choice for storing and retrieving unstructured image data. Also, Oracle is an expensive option.

- ☒ Store images in Amazon S3 buckets. Use Amazon DynamoDB with the geographic code as the key and the image S3 URL as the value.

This solution is highly scalable and cost-effective. S3 is designed for storing large objects like images, and DynamoDB is a NoSQL database that excels at handling high volumes of key-value lookups. The geographic code acts as the key for efficient image retrieval. This option effectively separates storage and metadata management, leveraging the strengths of each service.

- ⊗ Store images and geographic codes in an Amazon DynamoDB table. Configure DynamoDB Accelerator (DAX) during times of high load.

While DynamoDB is suitable for this use case, storing the images directly within DynamoDB may not be the best approach due to the size limitations of DynamoDB items (400KB). DAX is beneficial for read-heavy workloads but doesn't address the storage limitations of the images themselves. Also, storing images directly in DynamoDB will be more expensive than storing them in S3.

- ⊗ Store images in Amazon S3 buckets. Store geographic codes and image S3 URLs in an Oracle database running on an Amazon RDS Multi-AZ DB instance.

Although this option utilizes S3 for image storage, it still relies on Oracle RDS for metadata management. This approach isn't as cost-effective as using DynamoDB for metadata, especially considering the simple key-value nature of the geographic code-to-image URL mapping. RDS Multi-AZ provides high availability but doesn't necessarily improve write scalability during peak update events.

Overall Explanation:

The most cost-effective solution is to store the images in Amazon S3 and use Amazon DynamoDB to store the geographic code as a key and the corresponding S3 URL as the value. S3 provides scalable and cost-effective storage for the high-resolution images, while DynamoDB offers the necessary scalability and availability to handle tens of thousands of updates every few minutes. DynamoDB is a good fit for key-value lookups and is well-suited to track the relationship between geographic codes and image locations in S3, which is more cost effective than using a relational database such as Oracle RDS. Storing the images directly into DynamoDB isn't efficient due to size limitations and cost, and DAX primarily addresses read performance, not write scalability for a large image update scenario.

[View Discussions](#)[Ask AI](#)

Question 64

Not Attempted

Storage



A company has an application that collects data from IoT sensors on automobiles. The data is streamed and stored in Amazon S3 through Amazon Kinesis Data Firehose. The data produces trillions of S3 objects each year. Each morning, the company uses the data from the previous 30 days to retrain a suite of machine learning (ML) models. Four times each year, the company uses the data from the previous 12 months to perform analysis and train other ML models. The

data must be available with minimal delay for up to 1 year. After 1 year, the data must be retained for archival purposes. Which storage solution meets these requirements MOST cost-effectively?

- ☐ Use the S3 Intelligent-Tiering storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

S3 Intelligent-Tiering is for unpredictable access patterns, which is not the case here. The access patterns are known.

- ☐ Use the S3 Intelligent-Tiering storage class. Configure S3 Intelligent-Tiering to automatically move objects to S3 Glacier Deep Archive after 1 year.

S3 Intelligent-Tiering cannot be configured with a lifecycle policy to move directly to S3 Glacier Deep Archive.

- ☐ Use the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

Using S3 Standard-IA from the beginning will impact the cost. Data accessed every morning for 30 days should be stored in S3 Standard.

- ☒ Use the S3 Standard storage class. Create an S3 Lifecycle policy to transition objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days, and then to S3 Glacier Deep Archive after 1 year.

This option is the most cost-effective because it utilizes S3 Standard for the first 30 days of frequent access, transitions to S3 Standard-IA for infrequent access up to 1 year, and then archives to Glacier Deep Archive after 1 year for long-term archival.

Overall Explanation:

The most cost-effective solution is to use S3 Standard for the first 30 days when the data is accessed daily. After 30 days, the data is accessed quarterly, making S3 Standard-IA the appropriate storage class. After 1 year, the data is moved to S3 Glacier Deep Archive for archival purposes. This strategy minimizes storage costs while meeting the required access patterns.

 View Discussions

 Ask AI

[← Back to Exam](#)



Marketing

[Affiliate Program](#)

[Unlimited Access](#)

[Business Unlimited Access](#)

Support

[Contact Us](#)

team@itcertifications.io

Company

[Udemy Alternative](#)

[Whizlabs Alternative](#)

[TutorialsDojo Alternative](#)

Legal

[Terms of service](#)

[Privacy policy](#)

[Refund policy](#)

[Shipping policy](#)



© 2026, ITCertifications Technologies. All rights reserved.